



FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

**Institute for Factory Automation
and Production Systems**



Friedrich-Alexander-Universität
Technische Fakultät

Introduction in the production of electric machines

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

The lecture provides a practice-oriented overview of the main methods and technologies used in the production of electrical machines.

Electric drives are key components for the megatrends of modern society.

The production of electric drives involves a variety of complex processes, often with a high degree of manual labor.

After the lecture you will be able to

- Describe the basics of electrical drive technology,
- distinguish between the common designs of electric drives,
- select suitable manufacturing processes for electrical machines,
- and explain the manufacturing processes required in each case and the associated system technology.



The first lecture unit includes a general overview of electromechanical engineering and an introduction to the associated research area.

- **Organizational matters**
- Motivation and definition of terms
- Structure of an electrical machine
- Fundamentals of electromechanical energy conversion
- Structure of the fields of application
- Types, operation, and control systems of various electrical machines
 - Transformers
 - Commutator machines
 - Synchronous machines
 - Asynchronous machines
 - Other designs
- Current research projects



Graphic: BMW

The lecture is divided into a fundamentals and an advanced lecture.

"Fundamentals": Monday 12:15 pm - 1:45 pm. @H18				"Applications": Wednesday 2:00 pm - 4:00 pm. @FAPS-Egerlandstr. 7			
Date	Unit	Topic	Respons.	Date	Unit	Topic	Respons.
13.04.2026	F1	Introduction in the production of electric machines and Principles of electric motors and machines and their properties	Simon Stauber	15.04.2026	A1	Expert lecture FA Seamless Energy Technologies GmbH: Systems and production processes for wireless charging	Michael Masuch
20.04.2026	F2	Soft magnetic materials and manufacturing of lamination stacks	Ulf Schmidtgen	22.04.2026	A2	Expert lecture FA Stiefelmayer: Cutting of coils	Ulf Schmidtgen
27.04.2026	F3	Hard magnetic materials and rotor manufacturing	Thorsten Ihne	29.04.2026	A3	Excursion @ SIEMENS: Assembly of large motors	Felix Wirthmann
04.05.2026	A4	Power electronics production	Manuela Ockel	06.05.2026	F4	Winding technology: Wire materials and their processing	Valentin Henrich
11.05.2026	F5	Winding technology: Direct and indirect winding methods	Daniel Hößelbarth	13.05.2026	A5	Excursion @ tbd: tbd	Simon Stauber
18.05.2026	F6	Winding technology: Formed coil technology and its application in the automotive powertrain	Anja Preitschaft	20.05.2026	A6	Expert lecture FA Marsilli: Winding technology in practice	Valentin Henrich
25.05.2026	Whit Monday			27.05.2026	Bergkirchweih holiday		
01.06.2026	F7	Contacting technology	Miriam Eichinger	03.06.2026	A7	Excursion @ TESLA Automation: Hairpin stator production	Anja Preitschaft
08.06.2026	F8	Insulation systems and processes	Simon Stauber	10.06.2026	A8	Expert lecture FA Trumpf: Contacting technology	Simon Stauber
15.06.2026	F9	Mechanical components and assembly of electric machines	Alexander Vogel	17.06.2026	A9	Expert lecture FA Elantas: Insulation technology	Simon Stauber
22.06.2026	F10	Testing technology and quality control	Michael Masuch	24.06.2026	A10	Case study on winding and rotor design considerations	Valentin Henrich
29.06.2026	F11	Planning and optimization of production systems for electric motors	Benedikt Scheffler	01.07.2026	A11	Expert lecture tbd: tbd	Roman Hahn
06.07.2026	F12	Circular economy for electric machines	Thorsten Ihne	08.07.2026	A12	Analysis of the process chain of electric drives	Anja Preitschaft
13.07.2026	Backup date			15.07.2026	Backup date		

STIEFELMAYER
lasertechnik

SIEMENS

MARSILLI

Automation

TRUMPF

ELANTAS

The lecture is divided into a fundamentals and an advanced lecture.

From SS 2024, the exam 49501 "Elektromaschinenbau" will no longer be offered, but only the exam 49521 "Produktion elektrischer Motoren und Maschinen".

Fundamentals lecture

- Monday: 12:15 p.m. – 1:45 p.m.
- Room: H18
- The lecture slides are available for download on StudOn.
- The already uploaded script is the preliminary slides, which can be slightly improved until the Friday before the lecture.

Advanced lecture

- Wednesday: 2:00 p.m. - 4:00 p.m.
- FAPS seminar room, Institute FAPS, Egerlandstraße 7, 91058 Erlangen
- The course is divided into excursions with associated specialist lectures, specialist lectures as well as further in-depth lectures and exercises.
- The contents and documents stored on StudOn are relevant for the examination.
- Further information on the excursions and the registration forms can be found on StudOn.
- In order to enable as many people as possible to participate the excursions, we ask you to cancel your registration as early as possible if you are unable to attend.

All relevant documents for lectures and exercises are uploaded to StudOn.

studOn Link:

<https://www.studon.fau.de/cat2670013.html>

The screenshot shows the StudOn interface. At the top, there's a navigation bar with the StudOn logo and a search icon. Below it, a blue banner contains a notice: "BITTE BEACHTEN SIE: Während der vorlesungsfreien Zeit (12.02.-14.04.2024) reduzieren wir unsere Telefonzeiten. Die aktuellen Telefonzeiten finden Sie unter 'Support > StudOn-Support'." The main content area shows the course title "Produktion elektrischer Motoren und Maschinen (ehemals Elektromaschinenbau) (EIMB)" with a sub-header "Präsenzvorlesung über die Technologien für die Produktion von elektrischen Maschinen". On the left, there's a sidebar with icons for "Magazin", "Support", and "FAU Portale". Below the course title, there's a section "Kategorien" with a list of folders: "Altklausuren", "Vertiefende Inhalte zum Fach Elektromaschinenbau" (with a sub-note "Weiterführende Informationen und Medien"), "Elektromaschinenbau WS 2022/23" (with a sub-note "Vorlesungsunterlagen, Material, Links zu den Einzelveranstaltungen sowie Anmeldung zu Exkursionen und Veranstaltungen" and "Verfügbarkeit: 05. Okt 2022, 08:00 - 30. Apr 2024, 08:00"), "Elektromaschinenbau SS 2023 (Aufzeichnung)", and "Produktion elektrischer Motoren und Maschinen SS 2024". On the right, there's a blue button labeled "Aktionen".



The StudOn course "Produktion elektrischer Motoren und Maschinen - Grundlagen" is used. All content can be found here, including the advanced lectures. Registration for the "Applikationen" course is **neither necessary nor possible**.

The FAPS FAQs on StudOn answer most of the questions that arise.

Procedure for answering questions about the lecture and exercise

1. Read FAQs on StudOn

3. Write to the lecture or unit supervisor

Question was not answered

2. Post a question in the StudOn forum

Question was not answered

Organizational and content-related questions

Simon Stauber

simon.stauber@faps.fau.de

Contact for content-related questions

Corresponding supervisor of the respective lecture or exercise unit Contact details see:
<https://www.faps.fau.de/lehrstuhl/mitarbeiter/>

The examination is a written examination in German and lasts 60 minutes.

Examination

- 49521 "Produktion elektrischer Motoren und Maschinen"
- **Written** exam, **German** language
- 5 ETCS, **60 minutes**
- Registration via campo **within exam registration period!**

Exam contents

- Fundamentals lecture
 - Lecture slides
 - The spoken word of the lecturer
- Advanced lecture
 - Lecture slides provided on studOn
 - The spoken word of the lecturer
- Transfer questions

Transfer of the knowledge learned to practical tasks

Dates (organized by the examination office)

- Exam summer semester
 - Beginning of October (expected)
 - No individual dates for oral examinations
- Examination winter semester
 - End of March (expected)
 - Oral, if applicable
 - Content from the previous summer semester

Please note updates during the lecture!

Organizational matters

- Scope of examination
 - 120 Points
 - Examination time: 60 minutes
- Permitted accessories
 - Ballpoint pen
 - Pencil
 - Ruler
 - non-programmable calculator

As part of the advanced units, there will be an excursion to Siemens Mobility in Nuremberg.

Motivation

- Illustration of the production process chain for electric motors
- Linking teaching knowledge and the store floor
- Networking prospective engineers with local production companies

Key data on the excursion

- Date: Wednesday, April 29th, 2026
- Time: 2:00 p.m. – ~4:00 p.m.
- Company: **SIEMENS Mobility GmbH**
- Meeting Point: **Gugelstraße 76, 90459 Nürnberg**
- Binding Registration via StudOn:
<https://www.studon.fau.de/studon/go/crs/6898811>



Graphics: SIEMENS Mobility GmbH

SIEMENS



— STUD  N —

As part of the advanced units, there will be an excursion to Tesla Automation in Prüm by bus.

Motivation

- Illustration of automated processes for the production of hairpin stators
- Linking teaching knowledge and the store floor
- Networking prospective engineers with automation companies

Key data on the excursion

- Date: Wednesday, June 3rd, 2026
- Time: ~6:00 a.m. – ~9:00 p.m.
- Company: **Tesla Automation GmbH**, 54595 Prüm
- Meeting Point: **TechFak, 91058 Erlangen**
- Binding Registration via StudOn:
<https://www.studon.fau.de/studon/go/crs/6913649>

Grafics: Tesla Automation



Automation



The following literature can be used to reinforce the content of this lecture:

- Tzscheutschler, R., H. Olbrisch und W. Jordan. *Technologie des Elektromaschinenbaus*. Berlin: Verlag Technik, 1990. ISBN 3341008519
- Feldmann, K., V. Schöppner und G. Spur, Hg. *Handbuch Fügen, Handhaben, Montieren*. München: Carl Hanser Verlag GmbH & Co. KG, 2014. Handbuch der Fertigungstechnik. ISBN 978-3-446-42827-0
 - Dobroschke, A. Wickeltechnik. S. 197-221
 - Junker, S. Prozesse und Technologien für den Elektromaschinenbau. S. 751-769
- Hagedorn, J., F. Sell-Le Blanc und J. Fleischer. *Handbuch der Wickeltechnik für hocheffiziente Spulen und Motoren*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2016. ISBN 978-3-662-49209-3
- Kampker, A. *Elektromobilproduktion*. Berlin: Springer Vieweg, 2014. ISBN 978-3-642-42021-4
- Jordan, W. *Technologie kleiner Elektromaschinen Teil 1. Wissensspeicher für Produktentwickler, Techniker und Kaufleute*. 5. Aufl. Dresden: Eigenverl. technoexpert dresden, 2013. ISBN 978-3-00-039888-9

In addition, further specialist literature is provided in the respective set of slides for in-depth study and citation of the contents of the individual lecture contents.



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Graphic: BMW

Electrical machines are one of the most important forms of propulsion of the 21st century.

In order to understand the issues involved in the production of electrical machines, essential knowledge of the electrical engineering fundamentals of electrical machines and their areas of application is required.

After attending the lecture unit "Introduction in the production of electric machines" you should:

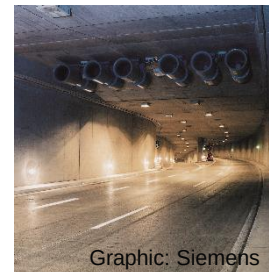
- **know the nomenclature of the most important components of electric motors,**
- **have understood the functional principle on which the electric machine is based**
- **have gained an overview of the main areas of application for electric motors and**
- **be able to describe the operating behavior of various electric machines,**
- **receive an overview of unconventional developments in electric machines.**



Graphic: Omron



Graphic: ZF Friedrichshafen

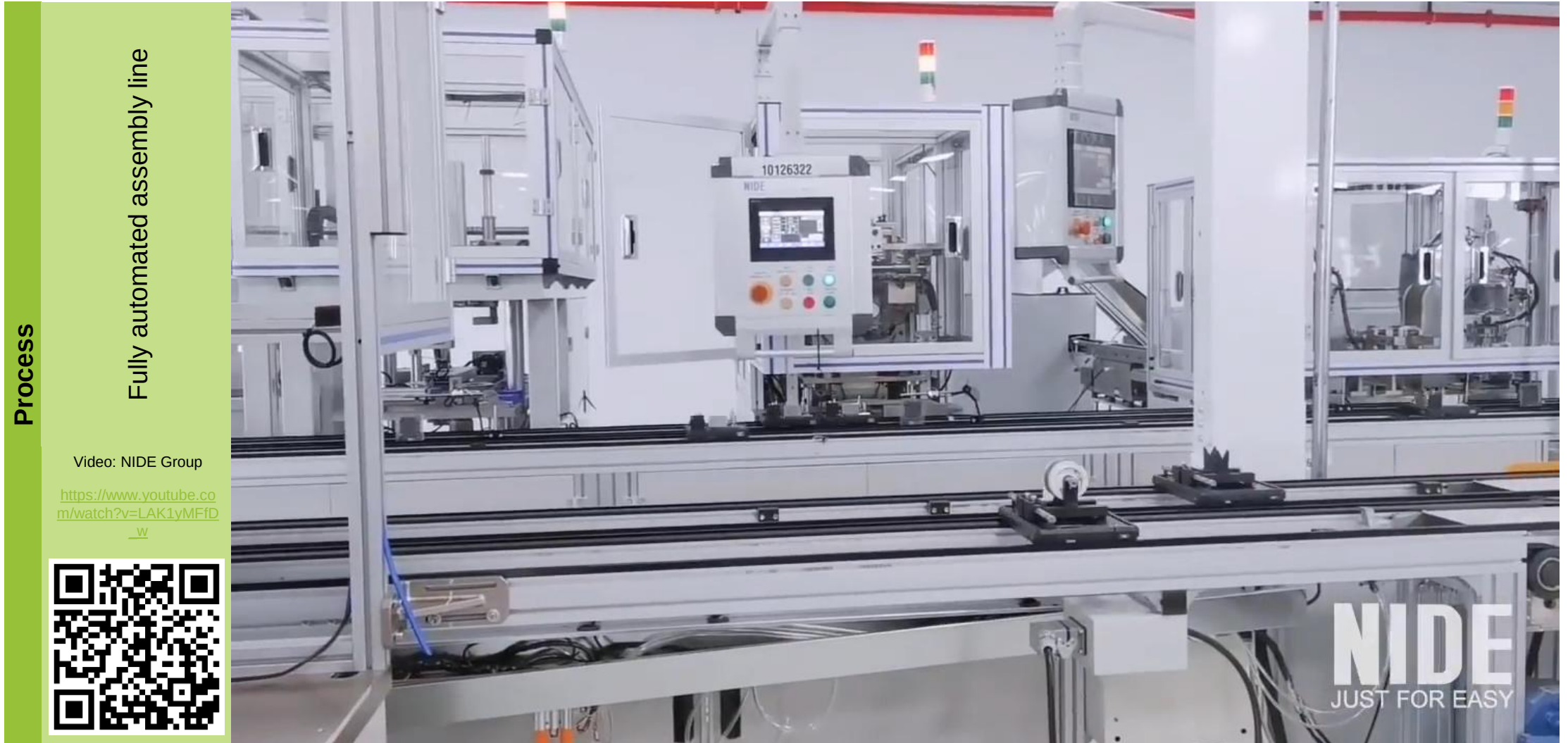


Graphic: Siemens

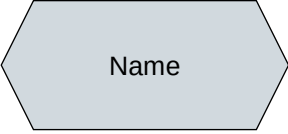
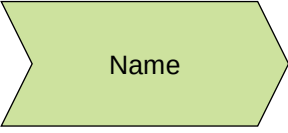
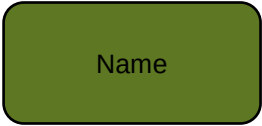
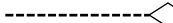
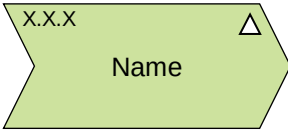
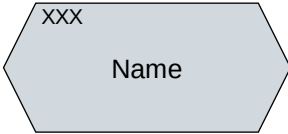


Graphic: Kuka

The joining of the two main assemblies of an electric motor, the rotor and stator unit, is the central joining operation of the final assembly.



The process chain is illustrated in the lecture based on the CONSENS notation elements.

	Elements	Relations	Extension
Partial model Processes	Material elements 	Sequence 	Logical operators: Exclusive Or  Inclusive Or 
	Processes 		
Partial model Resources	Resources 	Material flow   Assignment  Transport assignment	Numbering logic:  

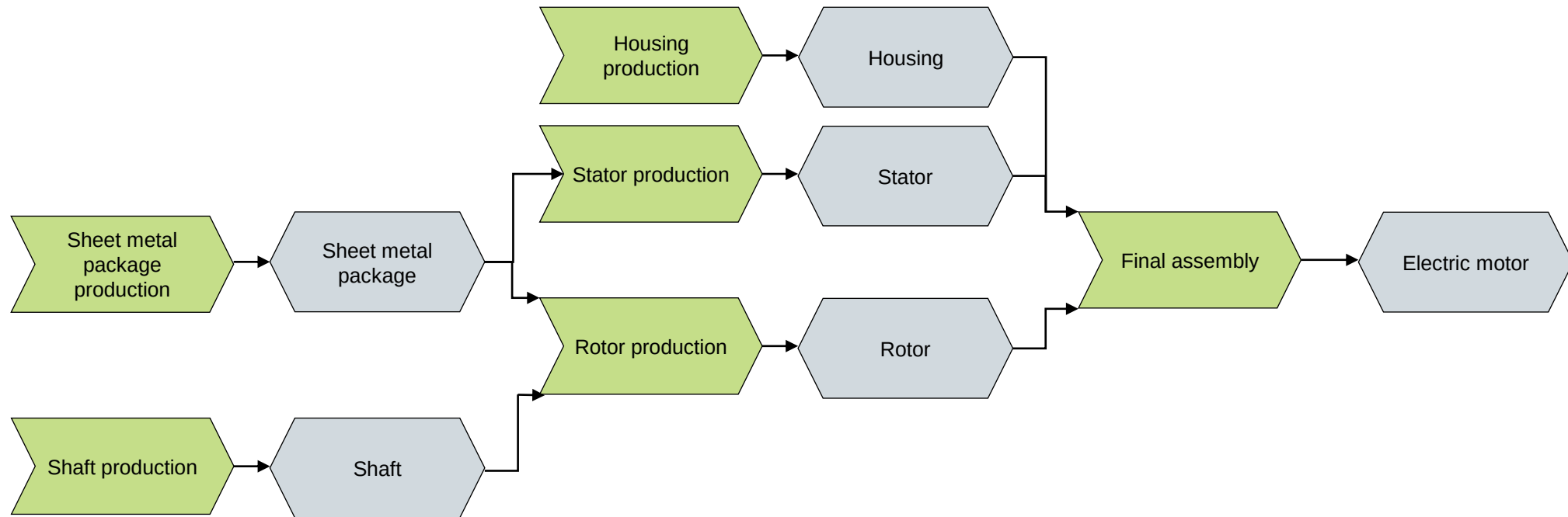
Numbering of processes:

- Sub-processes 1/2/3/4/5
- Sub-process steps 1.1 / 1.2 / 1.3
- Procedures: 1.1.1 / 1.1.2 / 1.1.3
- Alternatives are differentiated at all levels using letters in the numbering. If the alternatives have a different number of (several) steps, the steps are counted after the letter until the branches are merged. (1.1a1, 1.1a2, 1.1a3) If there is only one step, the number is omitted (at process level) 1.1.a, 1.1.b instead of 1.1.1a, 1.1.1b

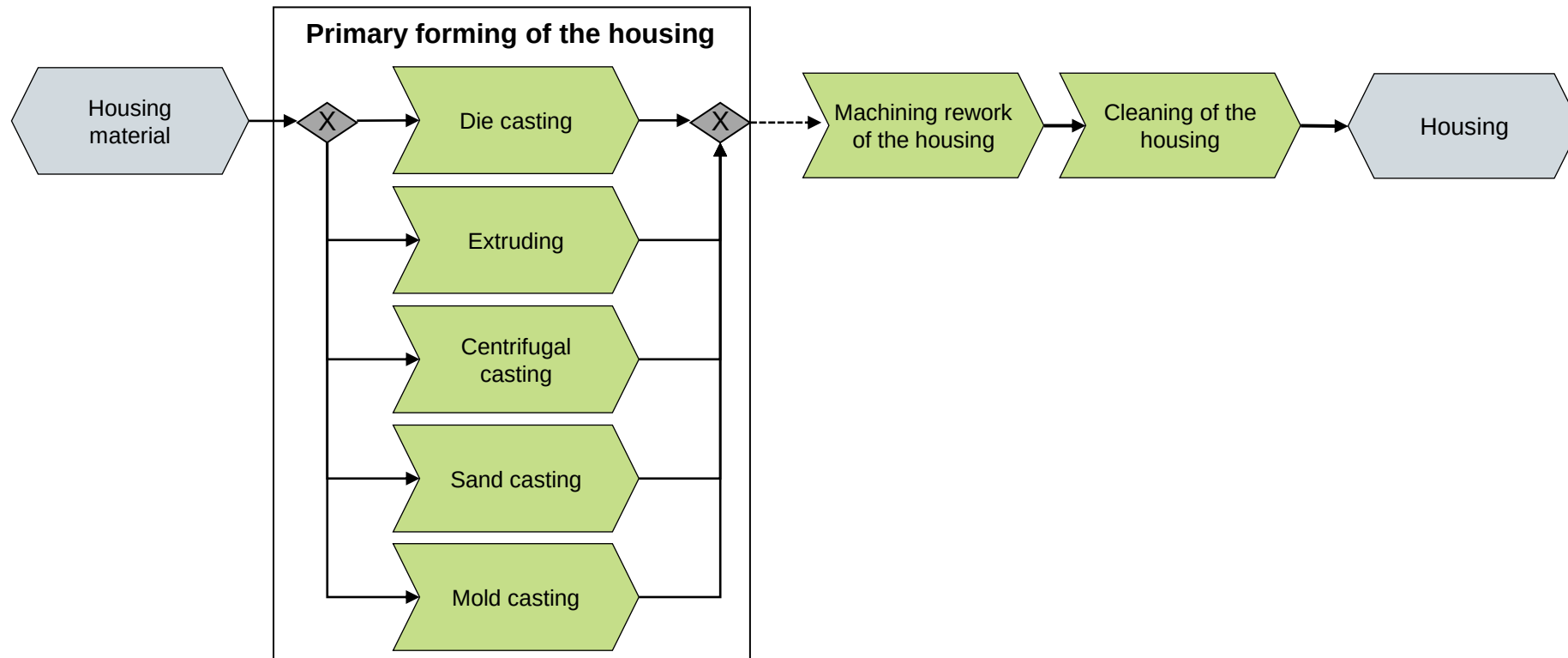
Numbering of material elements:

Main components according to the creation process, all other components numbered according to the main component. 1 Housing, 1.1 Housing raw material, 1.2 ... , 1.3 ... If there is a detailing of the general component at process level, this is counted through on the third level.

The process chain for manufacturing an electric motor comprises a large number of process steps and material elements.



Process alternatives are represented by excluding disjunctions. Various process alternatives are available for the primary forming of the motor housing.



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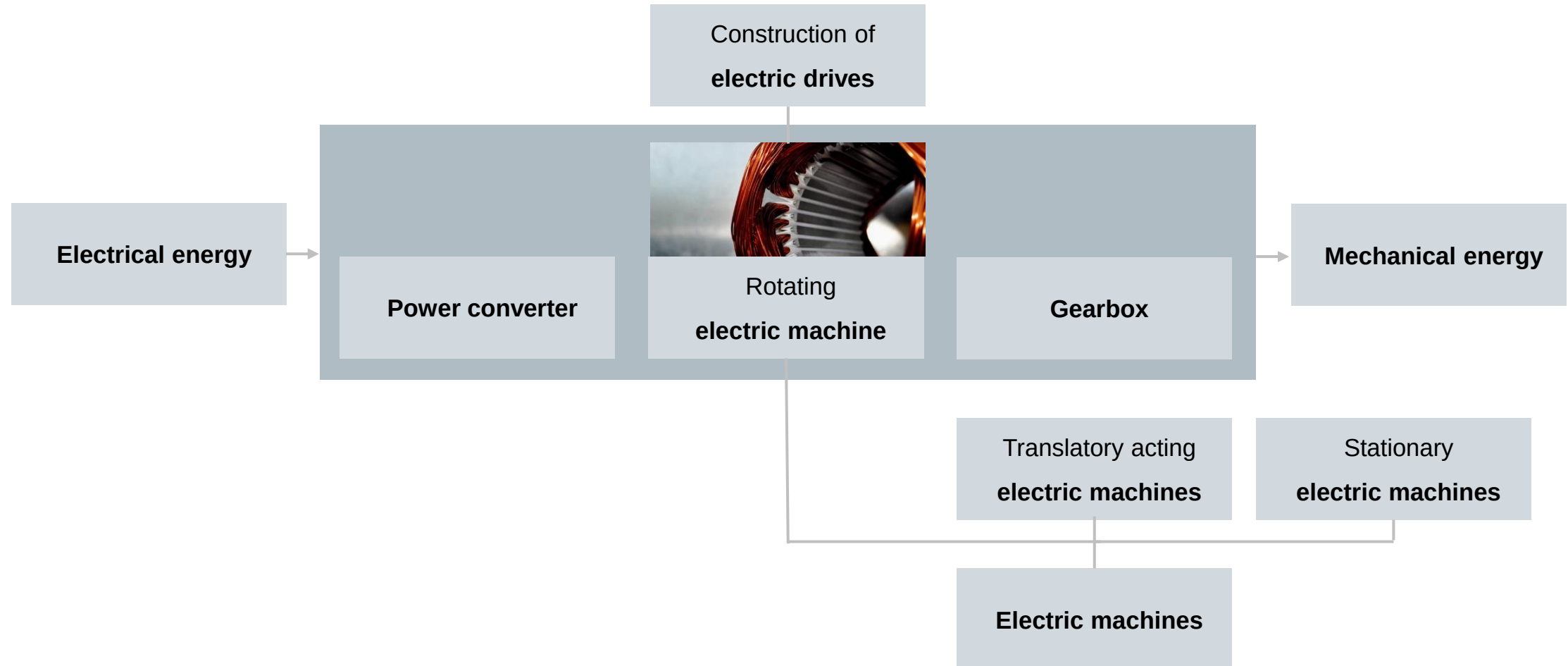
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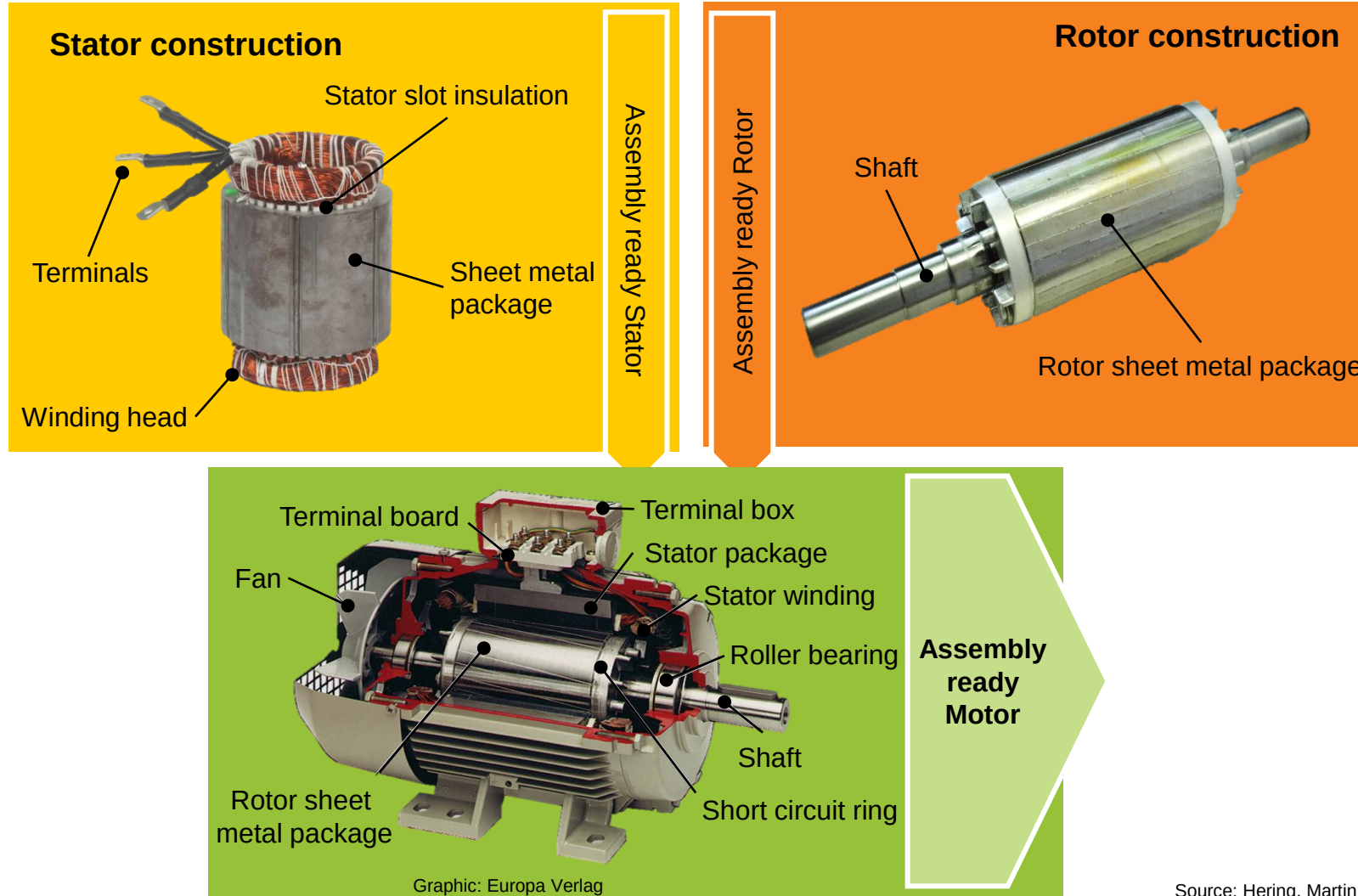
Graphic: BMW

An electric drive is an electromechanical energy converter. It consists of a functional unit that contains a power converter, an electrical machine and a gearbox.

Definition of terms



Rotating electric machines consist of a stationary part - the stator, and a rotating part - the rotor.



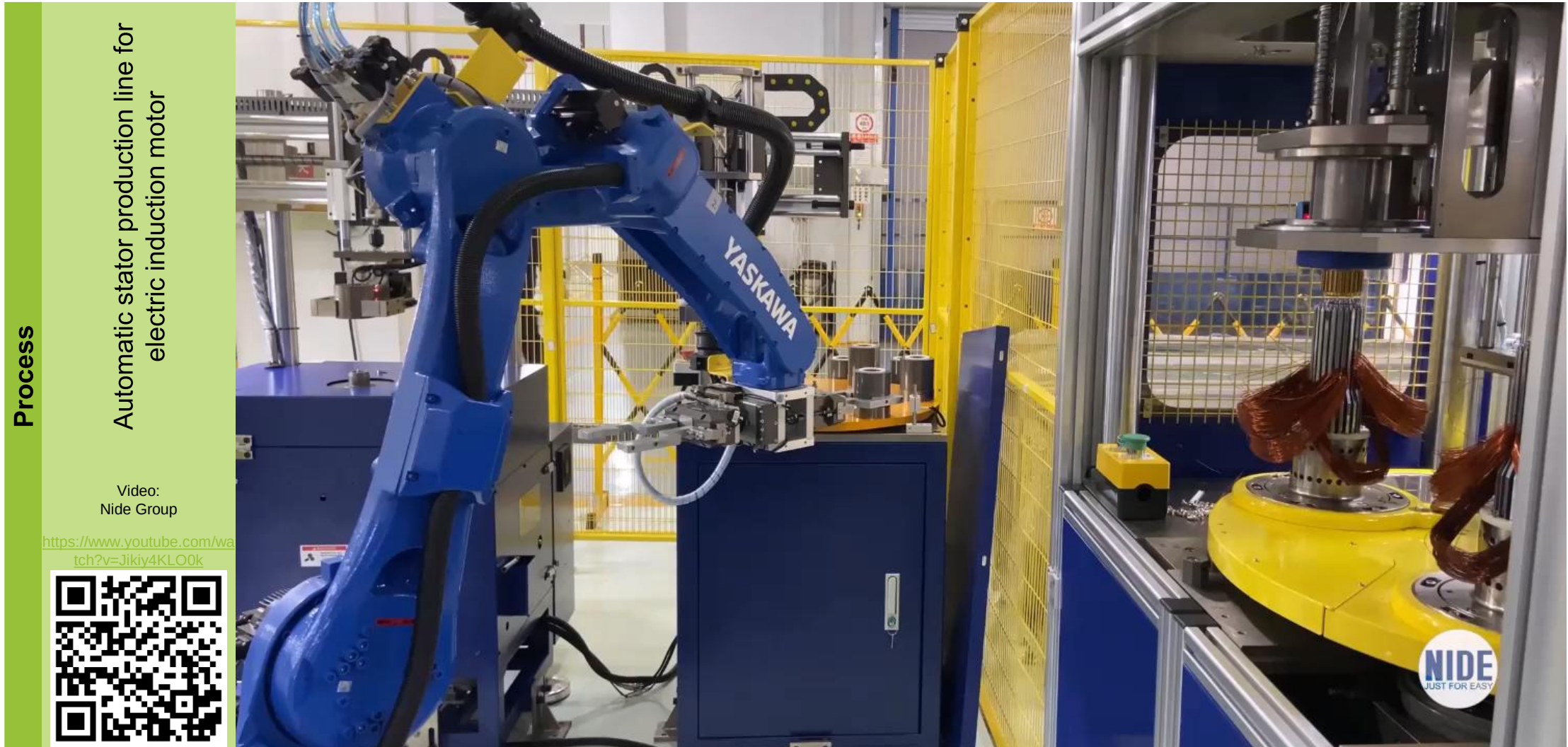
- The stator serves to create the rotating magnetic field.
- The rotor converts a magnetic moment into a mechanical torque and transfers power to the shaft.
- The housing serves to supply and protect the motor.

Source: Hering, Martin et al. 2012 – Elektrotechnik und Elektronik für Maschinenbauer

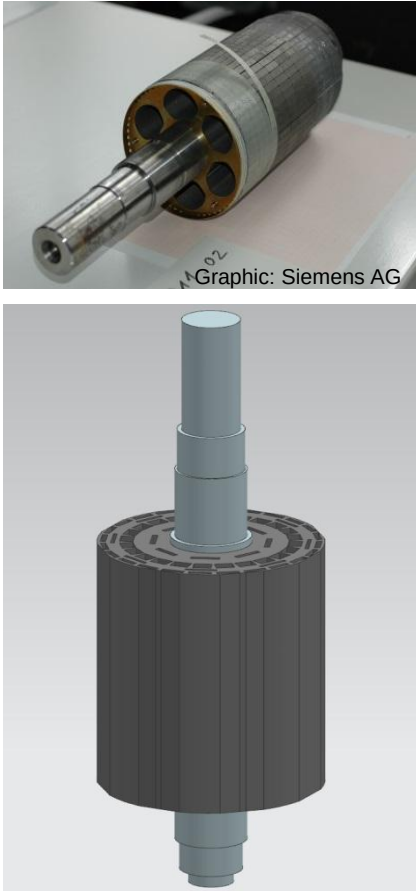
The stator serves to create the rotating magnetic field.

Assembly	Components	Physical function
  Graphic: Aumann GmbH  Graphic: EBMPapst GmbH & Co. KG	Stator	
	Winding wire	Create the electromagnetic field in the stator
	Enameled wire insulation	Create protective layer between individual wires
	Stator sheet metal package	■ Guide magnetic field ■ Transfer away heat produced in the windings
	Terminals	Evenly supply electrical power to stator coils
	Insulating resin	■ Protect stator windings from short circuits ■ Protect stator windings from external damage
	Stator slot insulation/ Slot closure	Create a protective layer between the electrical potential of the stator body and the electrical potential of the phase windings
	Winding head bandage	Mechanically fix the wires

The stator serves to create the rotating magnetic field.



The rotor converts a magnetic moment into a mechanical torque and transfers power to the shaft.

Assembly	Components	Physical function
 The image shows a photograph of a physical rotor assembly on the left and a 3D CAD model of a rotor on the right. The photograph shows a cylindrical metal component with a central shaft and a flange. The CAD model shows a similar cylindrical component with a central shaft and a flange, with a label 'Graphic: Siemens AG' at the bottom.	Rotor shaft	■ Mechanical transfer of torque ■ Ensure mechanical stability of the lamellar stack
	Rotor sheet metal package	■ Guide the magnetic field with minimal losses ■ In IPM-Motors: Protect magnets from mechanical damage
	Permanent magnets	Create magnetic poles required to generate mechanical torque
	Taping	■ Mechanically secure the magnets ■ Protect the magnets from mechanical damage
	End plates	■ Correction of imbalances ■ Stabilize the lamellar stack by creating axial compression
	Insulating resin/ Glue	■ Protect magnets from external influences (esp. corrosion) ■ Maintain magnets in fixed position (esp. high rotation speed)

The housing serves to supply and protect the motor.

Assembly	Components	Physical function
Housing	Housing box	- Protect motor from external influences - Heat removal from motor peripherals
	Bearings	Mechanically support rotor inside stator
	In some cases: Frequency converter	Modulation of amplitude, shape, and frequency of the electrical input signal
		
		

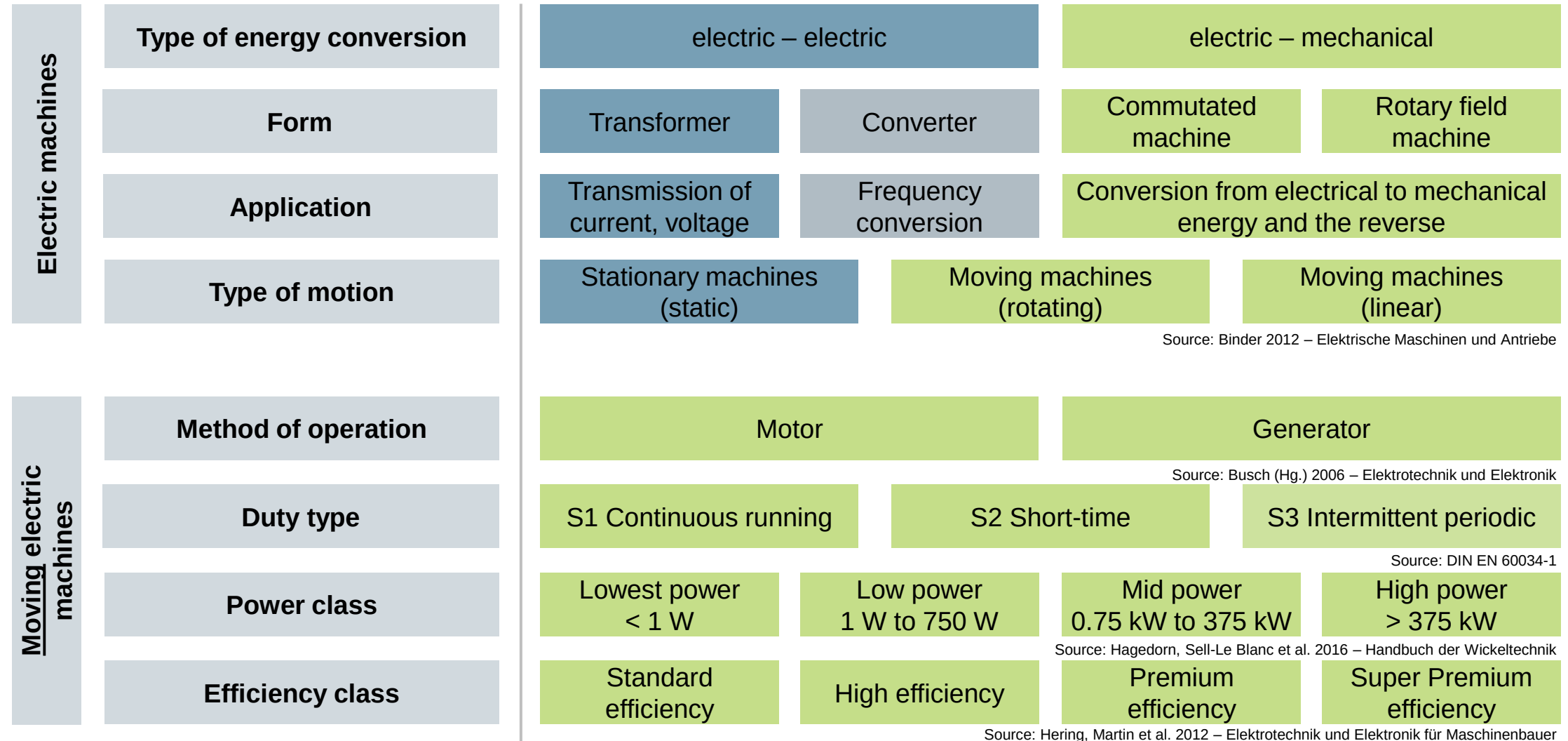
Graphic: Siemens

Graphic: Heinzmann GmbH & Co.

Graphic: Robert Bosch GmbH

Graphic: Robert Bosch GmbH

Electric machines may be categorized based on a wide range of criteria.



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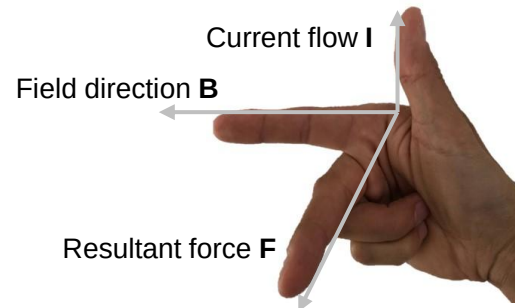
Graphic: BMW

The Lorentz force describes the effect of a magnetic field on a moving electric charge.

■ U-V-W-Rule (Right-Hand-Rule)

to determine the direction of the Lorentz force on a current-carrying conductor within a magnetic field:

- Cause: Technical current direction **I** from “+” to “-” (*thumb*)
- Influence: Field direction **B** from North to South pole (*index finger*)
- Effect: Resulting force **F** (*middle finger*)



■ Mathematical correlation in general via cross-product...

$$\vec{F} = L \cdot \vec{I} \times \vec{B} \text{ [N]}$$

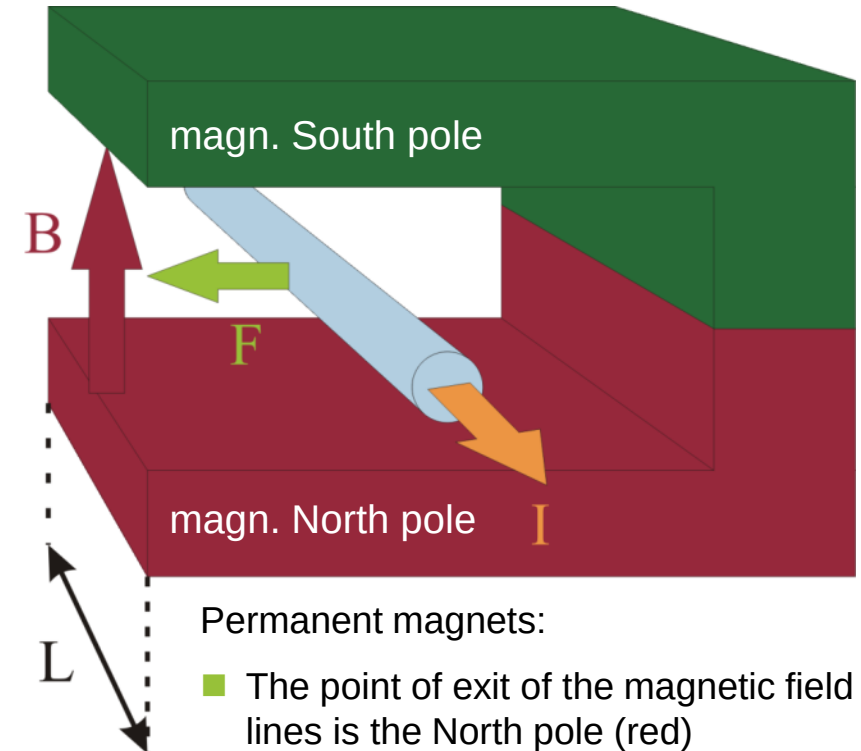
with:

- **F** – Resulting force on conductor [N]
- **L** – Length of conductor influenced by magnetic field [m]
- **B** – magnetic flux density [T]

■ Special case: perpendicular current flow direction (**I**) and B-field:

$$F = L \cdot I \cdot B \text{ [N]}$$

- With the help of the Lorentz force, the torque at the output shaft of an electric motor can be described. Increasing current or increasing number of windings increases the torque linearly.



Permanent magnets:

- The point of exit of the magnetic field lines is the North pole (red)
- The point of entry of the magnetic field lines is the South pole (green)

A loop of conductive material that is rotated in a constant magnetic field will cause a voltage to be induced at the terminals.

- The induced voltage U_{ind} , drives a current in the conductor loop whose magnetic field counteracts the apparent change in the external magnetic field according to Lenz's law:

$$U_{ind} = \frac{d\phi}{dt} [V]$$

- If an inductor with more than one turn is considered the flux linkage (ψ) of the entire coil may be calculated as follows:

$$\psi = n \cdot B \cdot A = n \cdot \phi [Wb]$$

with:

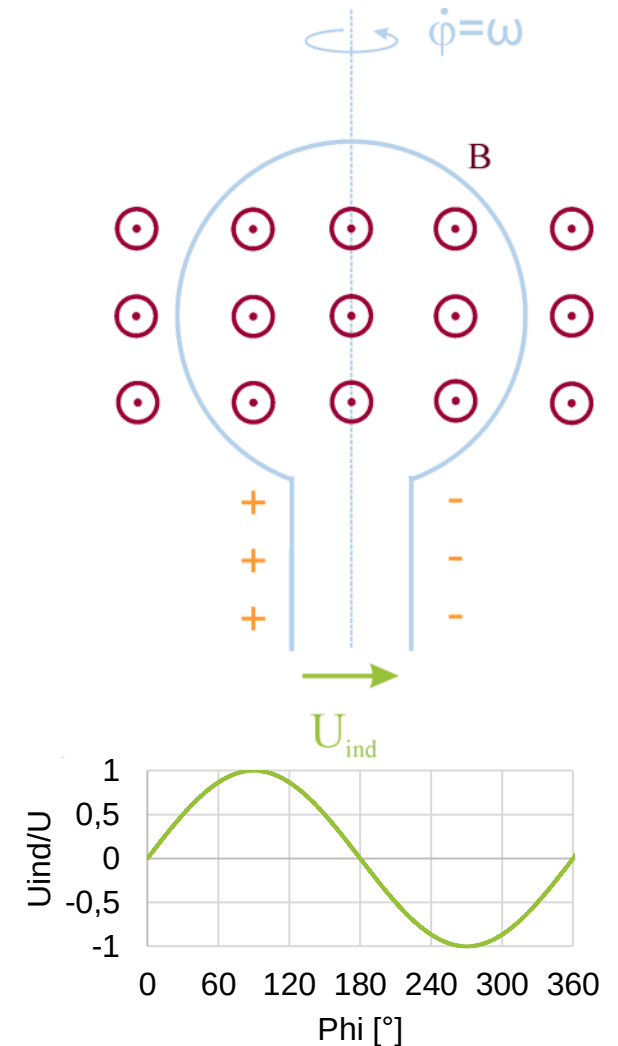
n – number of turns [unitless integer]

Φ – Magnetic flux [Wb]

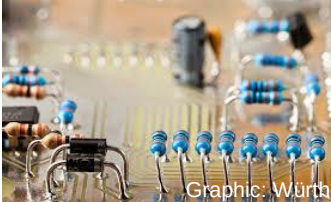
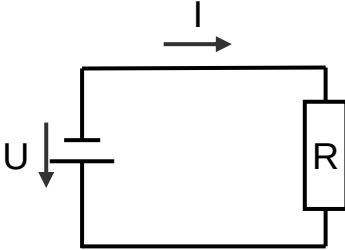
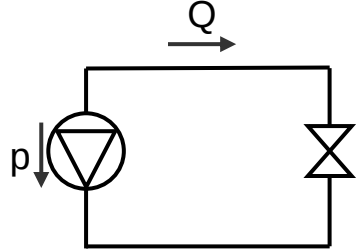
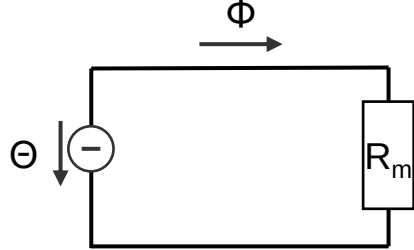
- The induced voltage is thus calculated according to:

$$U_{ind} = \frac{d\Psi}{dt} [V]$$

- More windings or higher flux density result in higher induced voltage for the same angular velocity ω .



Simplified calculation of magnetic circuits is based on electrical engineering using an equivalent circuit diagram.

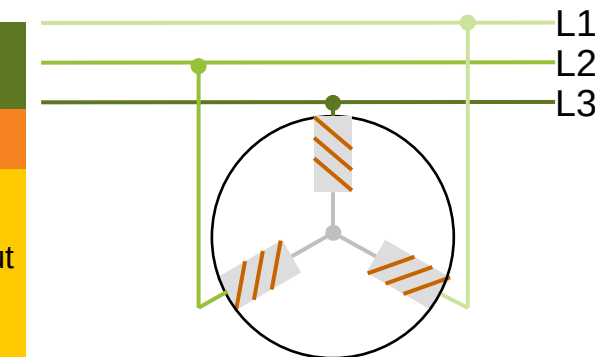
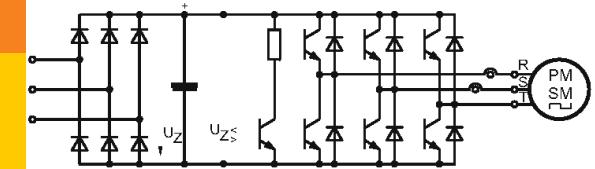
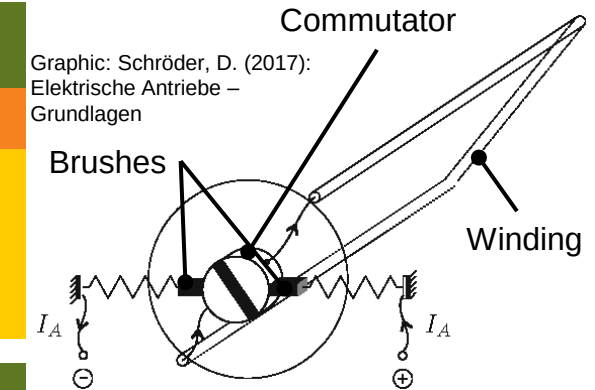
Electrical circuit		Hydraulic system	Magnetic circuit	
Component	Diagram	Diagram	Component	Diagram
 Graphic: Würth			 Graphic: Conrad	

Important parameters and analogous relationships

Electrical voltage U	↔	Electric field strength E	Pressure p	Magnetic excitation Θ	↔	Magnetic field strength H
Electric current I	↔	Current density S	Volumetric flowrate Q	Magnetic flux Φ	↔	Magnetic flux density B
Electrical resistance R			Back pressure/ Pressure drop	Magnetic Resistance R_m		

The classification of electric motors according to the type of current required for operation and the type of commutation results in three categories.

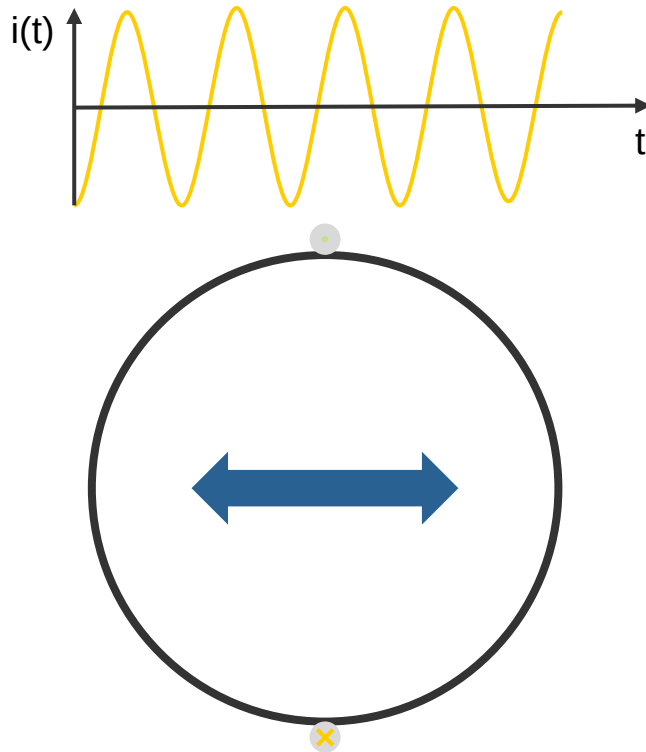
Direct current (DC)	Mechanical commutation	Commutator bars and brush contacts control armature current direction	
		Advantages - Easy to control - Good controllability via power converter - Excitation can affect characteristics	Disadvantages - Wear due to brush sparking and friction - Reversible pole- and compensation windings required to reduce brush wear (at $P > 30 \text{ kW}$)
Alternating current (AC), multiphase	Electric commutation	Control of (stator) current direction by regulated power converter	
		Advantages - Wear-free, no mechanical components - Independent control of frequency, current and voltage	Disadvantages (In SM) Rotor position must be determined - Complex power electronics required
	Direct operation with AC network	Control of (stator) current direction by 3-phase supply with sinusoidal AC voltage	
		Advantages - Wear-free (for ASM and PMSM) - Easy to control	Disadvantages - Drive speed is not variable - Synchronous motor does not start without starter device



Three-phase alternating current can be used to generate a rotating magnetic field.

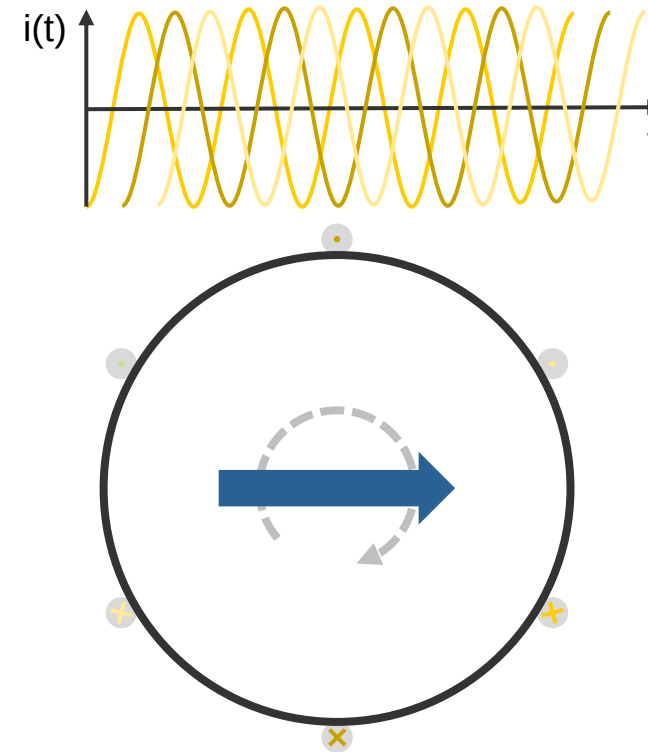
1-phase, 1 winding

- Sinusoidal current supply
- B-field scales in intensity and alternates direction



3-phase, 3 windings

- Separate sinusoidal current supplied to each phase with constant phase offset
- Coils are at the same angular offset as current phases (120°)
- B-field rotates



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Graphic: BMW

The branch of electric drives has the highest impact in Bavaria.



- Electric drive technology, with a € 4.5 billion share of production in 2016, was the largest branch of mechanical engineering in Bavaria.
- Over 200 Bavarian companies engage directly with the development, manufacture, and application of components and systems for electric drive technology.

Source: VDMA Bayern-Geschäftsbericht 2017

Electric drives are a core component of modern industrial products.

Material transport



Graphic: Jungheinrich

Automotive tech.



Graphic: Continental

Medical tech.



Graphic: Siemens

Machine tools



Graphic: Weiler

Transportation tech.



Graphic: SNCF

Micro drives



Graphic: FHG

1 mm³

Miniature drives



Graphic: Maxon

1 cm³

Geared motors



Graphic: ABM

1 dm³

1 m³

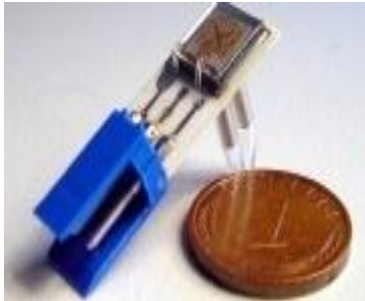
Large scale motors



Graphic: Siemens

Scale

Medical tech.



Graphic: Siemens

Automation



Graphic: KUKA

Household devices



Graphic: BSH

Plant engineering



Graphic: Sipsos

Energy tech.

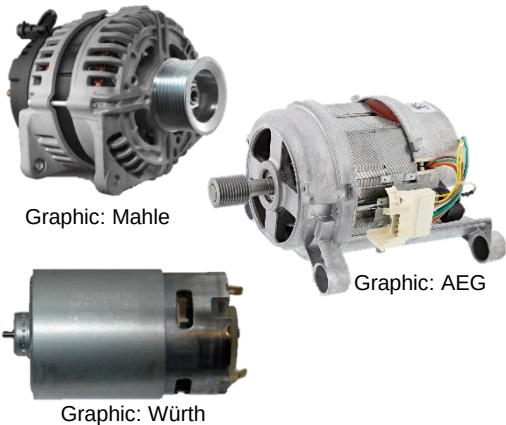


Graphic: Siemens

The application fields of electric drives may be classified into small scale, industrial, energy generation, and traction drives.

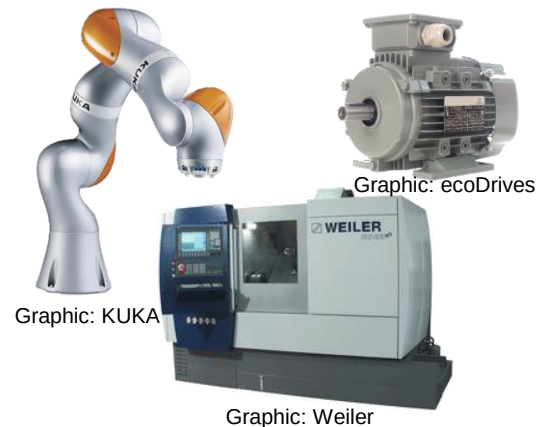
Small scale

- Household devices such as, washing machines, vacuum cleaners, refrigerators, mixers
- Consumer goods such as, printers, cameras, Blu-ray players
- Electric tools such as, drills, saws, garden tools



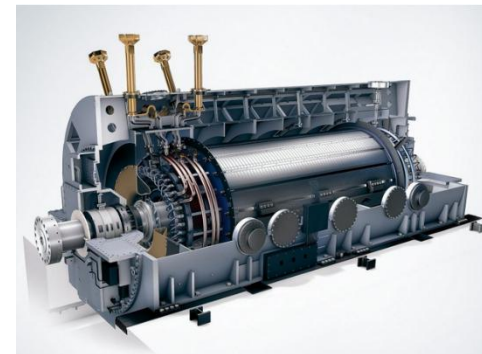
Industrial

- Continuous flow machines (Pumps, blowers, compressors)
- Machine tools (main and support drives)
- Robots (industrial, medical, or service)
- Conveyor drives (lifts, cranes, industrial tracks)



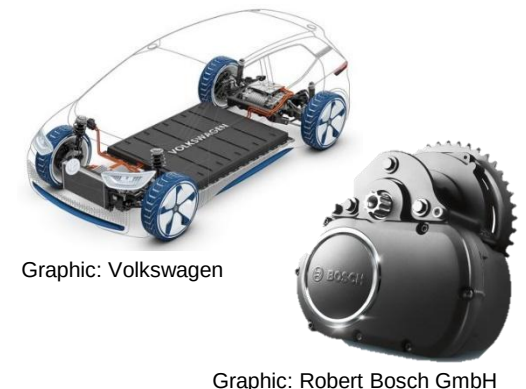
Energy generation

- Thermal power plants (coal, natural gas, nuclear)
- Wind power plants (wind turbines)
- Hydroelectric power plants (run-of-river, storage and tidal power plants)



Traction drives

- Low power traction drives (Scooter, E-Bikes)
- Mid power traction drives (Electric vehicles / -motorcycles, automobiles, -busses and trucks)
- High power traction drives (Ships, traction trains, dump trucks)



Product and manufacturing requirements for electric drives differentiate themselves according to their application field.

	Small scale	Industrial	Energy generation	Traction drives
Product requirements	■ Service life appx. 10 or more years ■ No maintenance requirements, consumable product ■ Low power requirements, up to appx. 2 kW ■ Power supplied through battery or consumer electric grid	■ Depreciation over appx. 10 years, often longer in operation ■ Maintenance free or maintenance friendly ■ Power output from a few Watt to several kW ■ Power supplied through electric grid ■ Energy efficiency often required by law	■ Service life 16 – 30 years ■ Highest possible ease of maintenance, often due to limited accessibility ■ Maximum size and weight defined by transportation requirements ■ Power output in the range of several MW ■ High energy efficiency	■ Service life of at least 15 years / 8.000 hours ■ As maintenance free as possible ■ Lowest possible size and weight with maximal power output ■ High energy efficiency to maximize range
Manufacturing requirements	■ Very high unit numbers, mass production ■ Minimal production cost	■ Various unit numbers: standardized products and customer specific products ■ Low production and operating costs	■ Low unit production ■ Lowest possible production costs	■ Moderate unit production, high growth expected ■ Lowest possible production costs

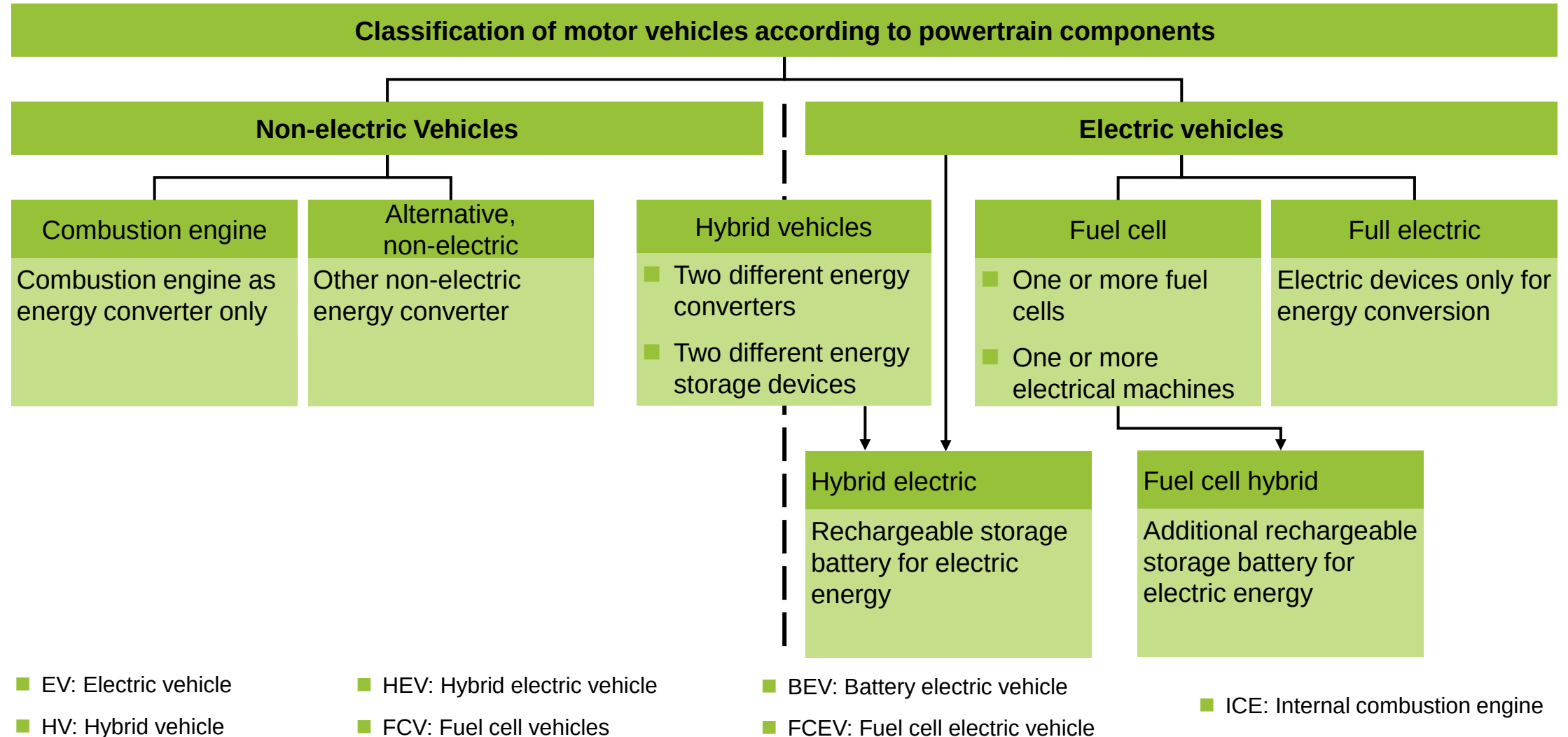
Source: Garbrecht 2008 – Auswahl von Elektromotoren

Electric cars have evolved from early 19th-century experiments to 21st-century mainstream adoption, driven by technological advances, environmental concerns, and supportive policies.

- 1828: First car with an electric motor is introduced
- 1837: First American patent is filled for an electric machine/motor
- 1859: The first EV with a lead-acid battery is launched
- 1884: Thomas Parker builds the first practical production EV with rechargeable batteries
- 1911: First gasoline-electric hybrid car is launched
- 1972: The first electric car from BMW was used at the 1972 Olympics.
- 1977: AMC and Gulton Industries launch a 3-passenger electric commuter city car
- 1996: General Motors introduces the EV1
- 1997: Toyota and Ford release the RAV4 EV & Ranger EV respectively
- 1998: Honda released the EV Plus
- 2001: Mahindra launches REVA, the first Indian EV
- 2003: Tesla motors is founded
- 2009:
 - BYD releases the world's first plug-in hybrid compact sedan F3DM
 - BMW starts initial trial of Mini Electric
 - Tesla unveils the Model S electric sedan,
 - U.S. Department of Energy awards US\$8 bn in loans to Ford, Tesla and NISSAN for development of EVs

Source: Statista

Electrifying the automotive powertrain occurs in a variety of ways, as defined by the United Nations.



Source: United Nations Economic and Social Council

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Graphic: BMW

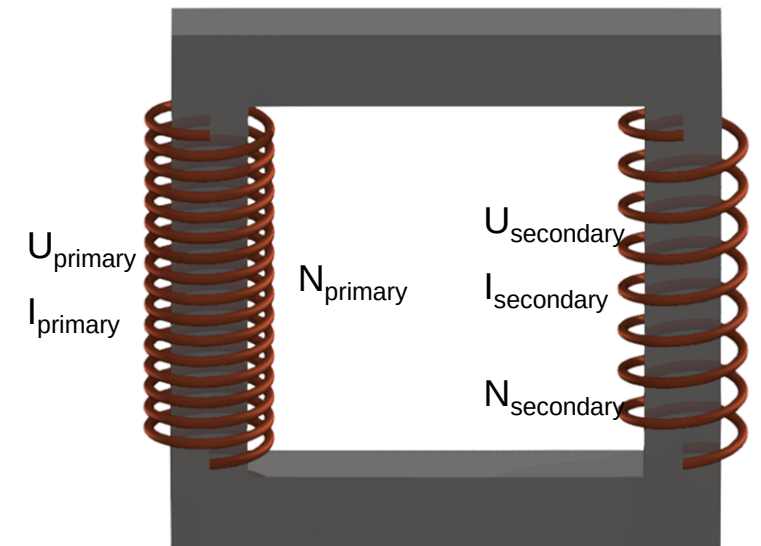
A transformer is capable of converting different electrical quantities.

Principles:

- The transformer is a stationary electrical machine, with no moving mechanical components
- Primary and secondary windings (N) are electrically separated from each other
- The magnetic coupling is achieved through a magnetically permeable core

Applications:

- Current and voltage conversion in distribution supply networks (also overland), power supply in general
- Resistance conversion and impedance matching for signal transmission
- Circuit protection (Isolation transformer, transformer ratio 1:1)

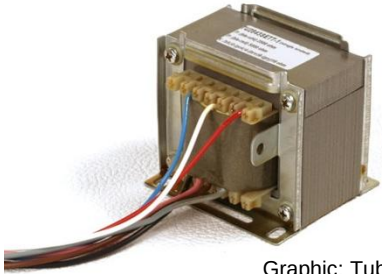


Ideal transformer conversion ratios

Voltage	Current	Resistance
$\frac{U_P}{U_S} = \frac{N_P}{N_S}$	$\frac{I_S}{I_P} = \frac{N_P}{N_S}$	$\frac{Z_P}{Z_S} = \frac{N_P^2}{N_S^2}$

In many transmission applications,
the transformer is used in various designs.

Audio output converter



Graphic: Tubetown

SMD Pulse transformer



Graphic: TDK

Mains transformer



Graphic: FAPS

Power transformer



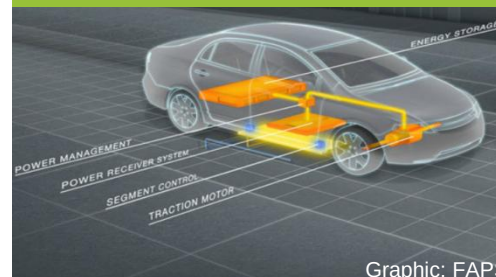
Graphic: ABB

Column variable transformer



Graphic: Ruhrstrat

Inductive charging
(air-gap transformer)



Graphic: FAPS

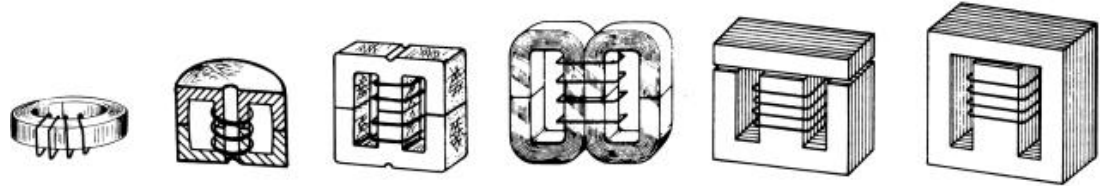
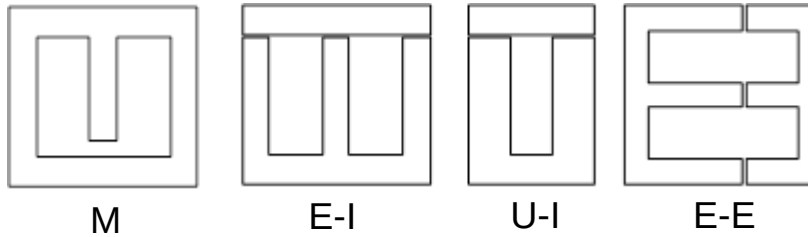
Ring variable transformer



Graphic: Ruhrstrat

Transformers can be constructed in different geometries to suit their intended application.

Possible designs of the lamellar stacked core:



Graphic: Böhmer E., Ehrhardt D., Oberschelp W. (2018) Transformatoren und Übertrager

Toroidal transformer

- galvanic isolation of the phases
- compact design
- at least 2 windings
- low stray magnetic field

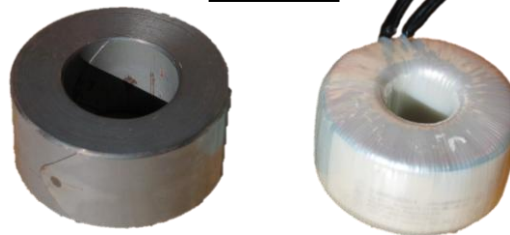
Three-phase AC transformer

conventionally most commonly used

Variable transformer

- for small outputs
- adjustable
- use in laboratory equipment

Toroidal



Radial sheet metal core (wound)

Wrapped and secured

Graphics: Wikipedia

Three-phase AC



Graphic: trafomodern

Variable



Graphic: Wikipedia

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Graphic: BMW

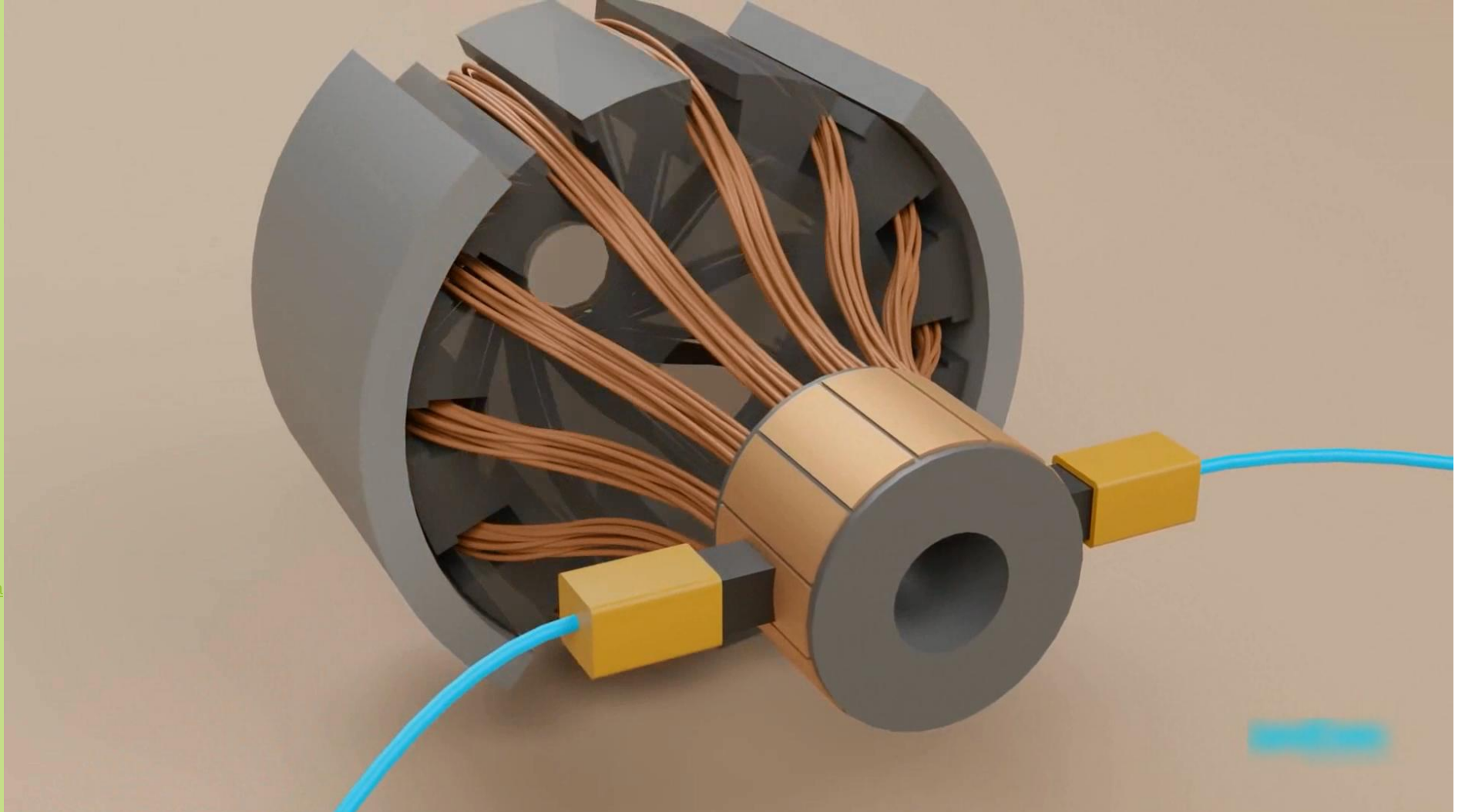
In order to generate rotation, continuous reversal of the rotor magnetic field at the correct time is required.

Working principle

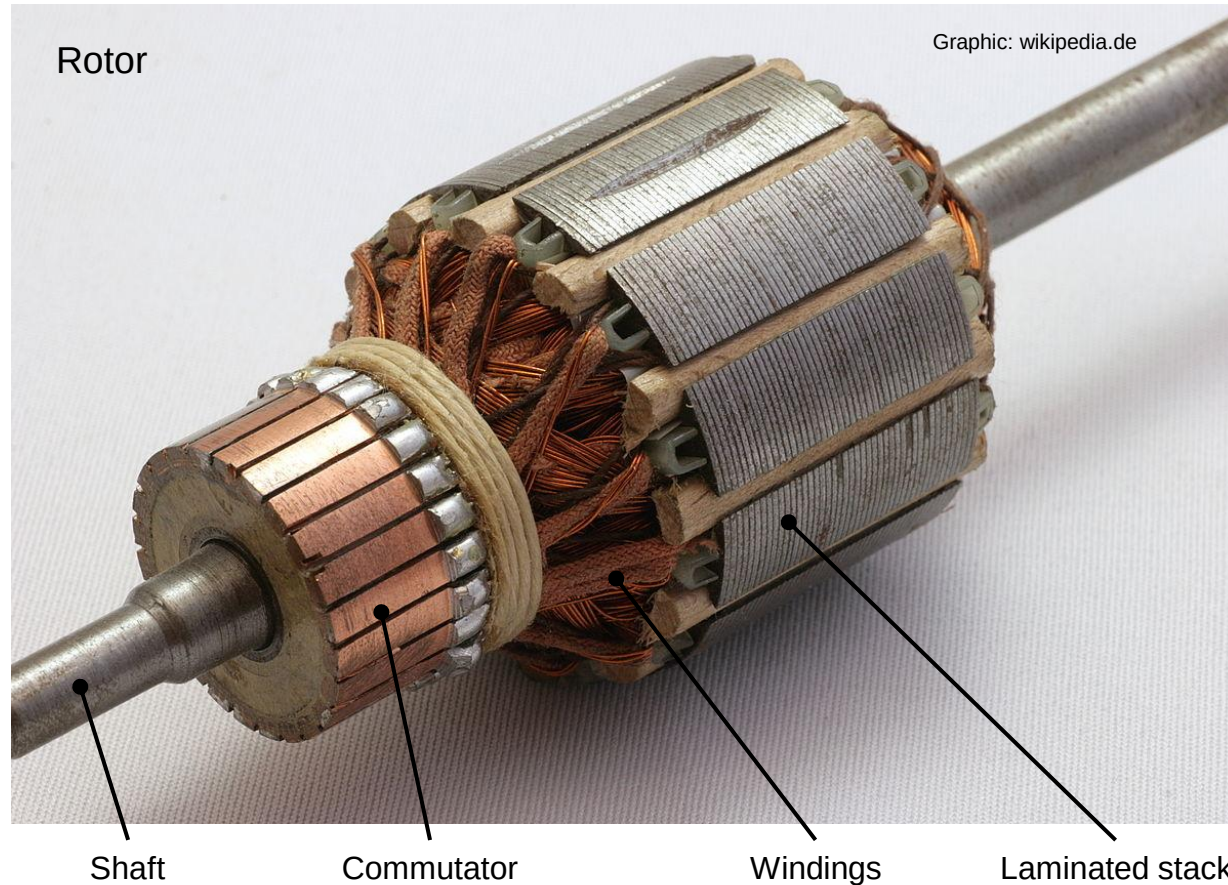
Commutator/DC machine

Video:
Jared Owen

<https://www.youtube.com/watch?v=CWuIQ1ZSE3c>



Due to many disadvantages, the mechanically commutated DC machine is increasingly being replaced by electronically commutated machines



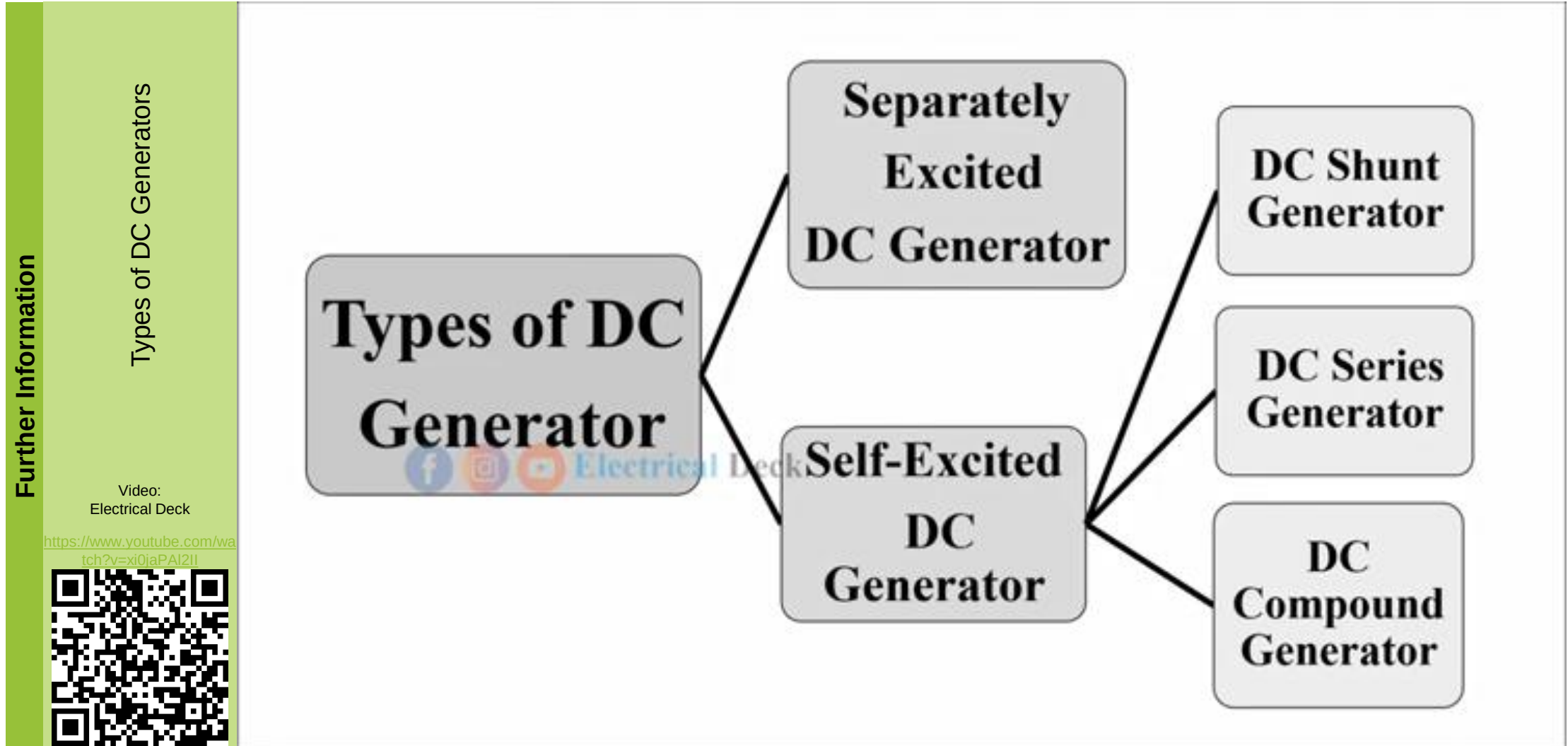
Advantages

- Ideally suited for DC applications
- DCM can be operated at variable speed with simple regulator systems
- With the appropriate electronics, the DCM can also be used as a servo drive
- Speed control possible by pulse width modulation

Disadvantages

- DCM is generally more expensive (commutator, rotor circuitry, reversible pole winding, compensating winding)
- DCM is more wear-intensive (brush sparking)
- The currently available power electronics (and low price) enable brush-free commutation

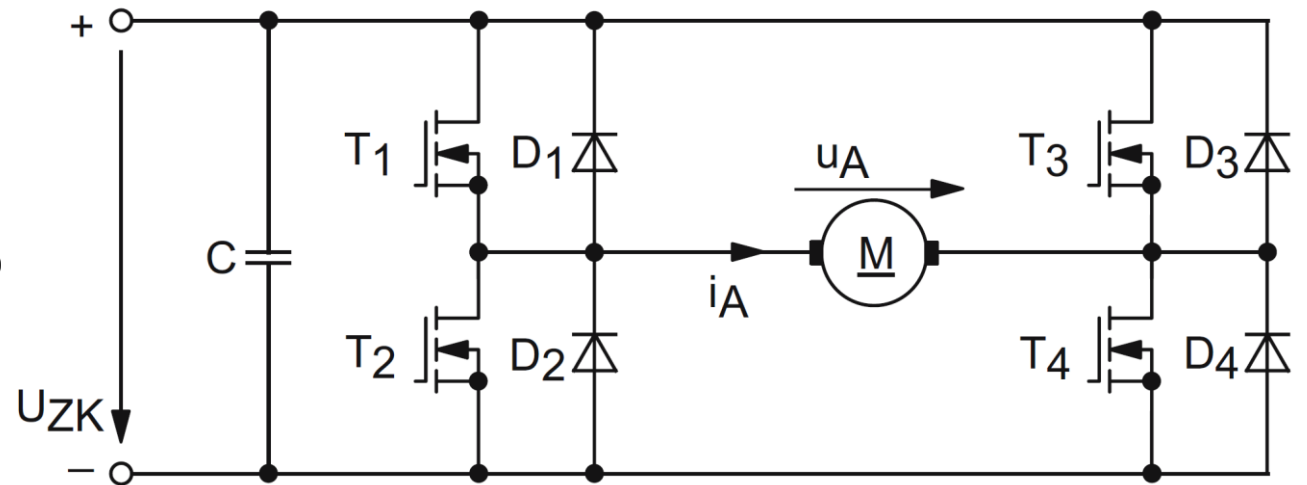
The type of excitation determines the torque-speed characteristic curve of the commutator machine.



DC machines can be efficiently supplied at variable speeds using various power converters.

Control via H-bridge transistor circuit:

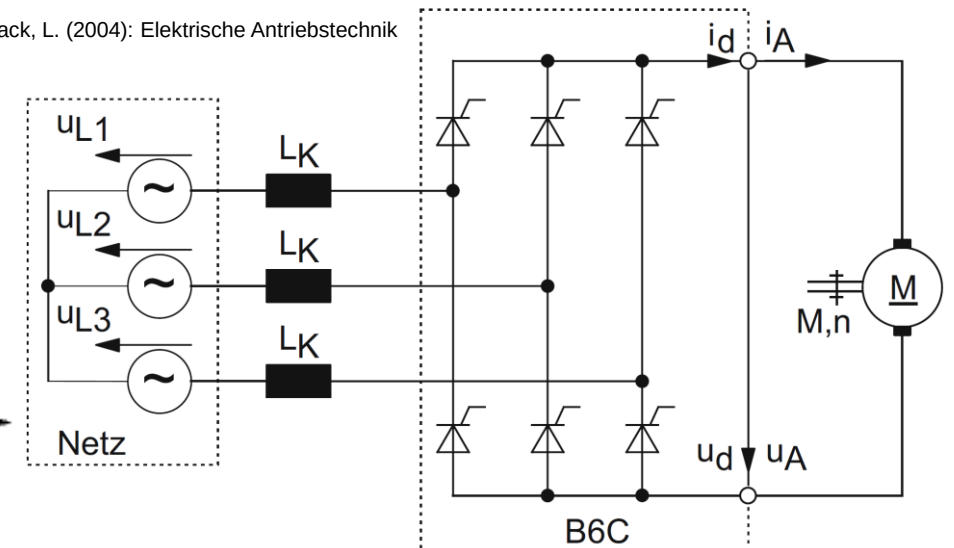
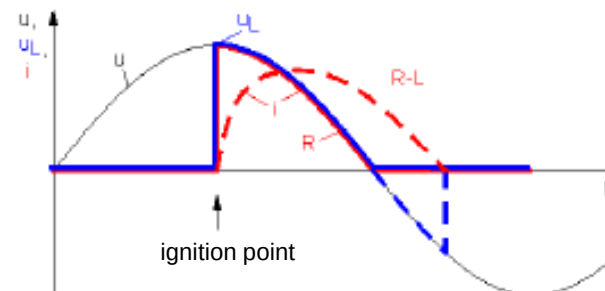
- Transistor T1 and T4 on: e.g. Right-hand rotation
- Transistor T3 and T2 on: Left-hand rotation
- To control switching pulses, one transistor (T1 or T2 and T3 or T4) is switched off per half-bridge (diode to eliminate flyback in the inductive load)



Graphic: Sack, L. (2004): Elektrische Antriebstechnik

Control via thyristor circuit B6C:

- Power supplied via 3-phase AC
- 6 thyristors are "fired" with the help of the phase connection control
- Thyristors equalize voltage
- The "firing angle" determines the level of the output voltage
- Mains-connected power converter



The range of applications for commutated machines ranges from small drives to large motors with outputs of up to 10 MW.

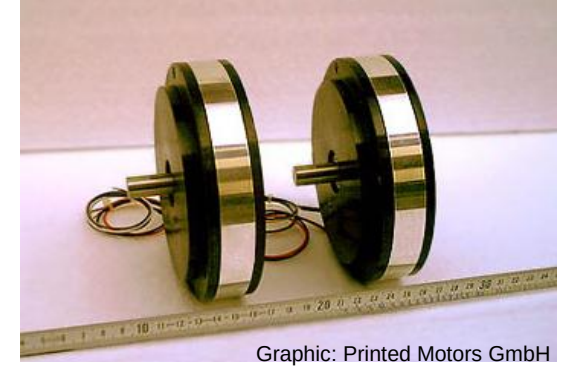
Large drives in mining equipment



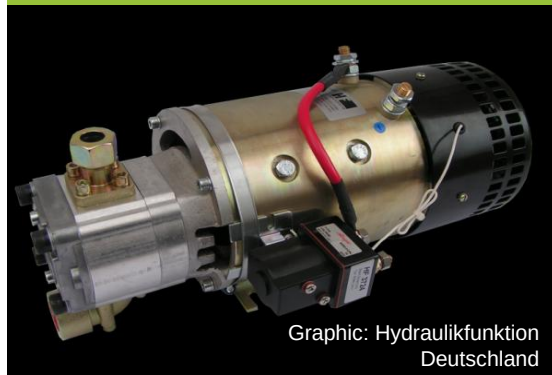
Large drives in rolling mills



Disc motor used as servo drive



Hydraulic pumps



Drives for revolving doors



Miscellaneous small drives



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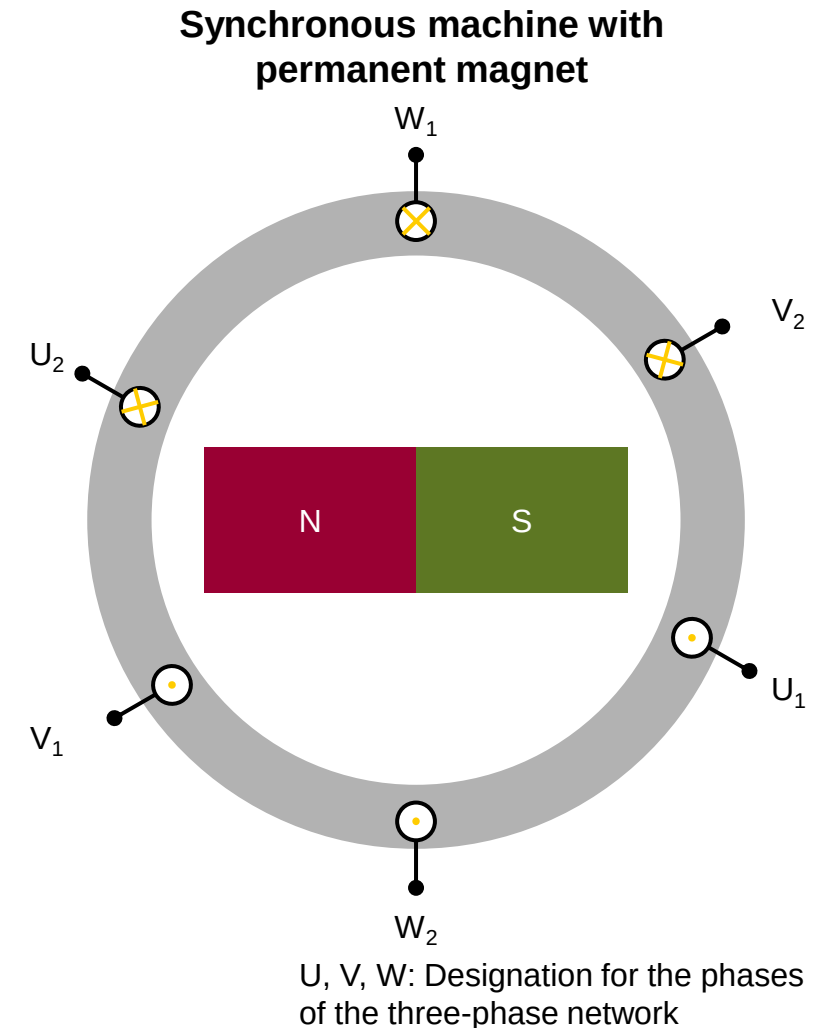
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Graphic: BMW

In the case of the synchronous machines, the rotor rotates synchronously with the rotating stator magnetic field.

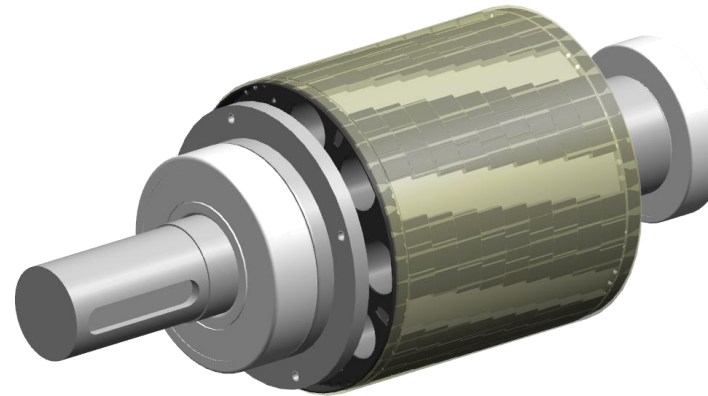
- The windings are usually housed in the stator.
- The stator windings are offset by 120° (in a two-pole motor).
- The stator is constructed of laminated sheets to reduce eddy currents.
- The (permanently excited) rotor aligns itself according to the B-field generated by the stator ("right-hand-rule" for B-Field).
- The position of the rotor must be detected in the case of variable-speed control for a targeted torque build-up.



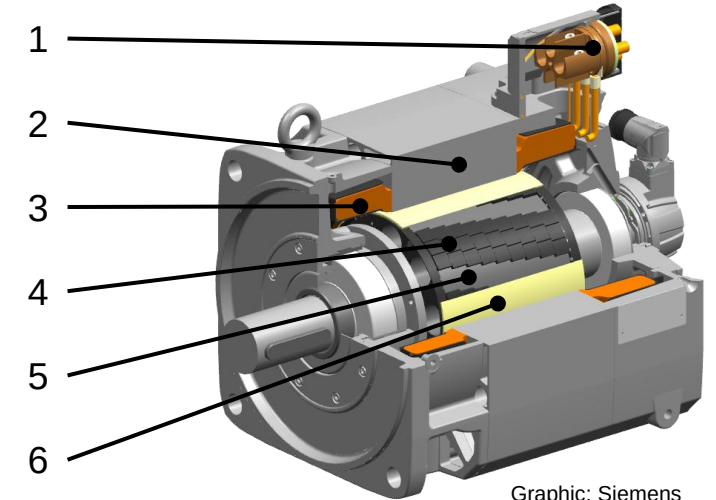
The permanently excited synchronous machine may be constructed using an external or internal rotor configuration.

Internal rotor-Synchronous machine with surface-mounted permanent magnets

- 1) Electric contacts
- 2) Stator lamination stack
- 3) Windings
- 4) Permanent magnets
- 5) Rotor lamination stack
- 6) Retaining sleeve



Graphic: Siemens

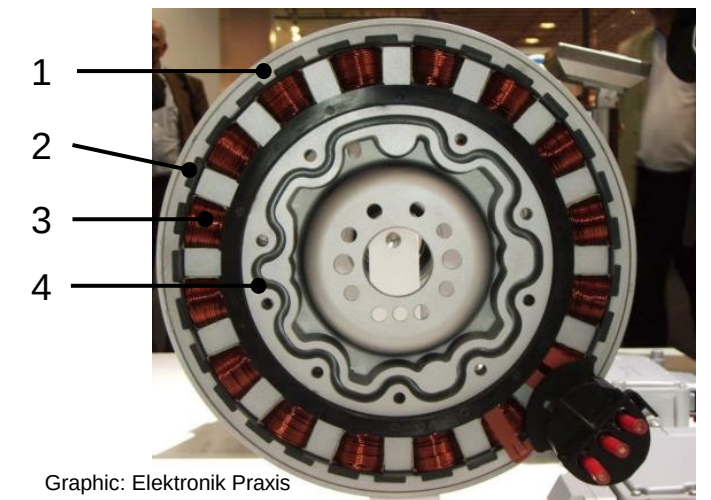


Graphic: Siemens

External rotor-Synchronous machine with surface-mounted permanent magnets

- 1) Rotor lamination stack with rotor carrier
- 2) Permanent magnets
- 3) (Concentric) windings (single-tooth windings) on the stator lamination stack
- 4) Stator carrier

The motor shown is used as a starter-generator in passenger cars and runs at crankshaft speed

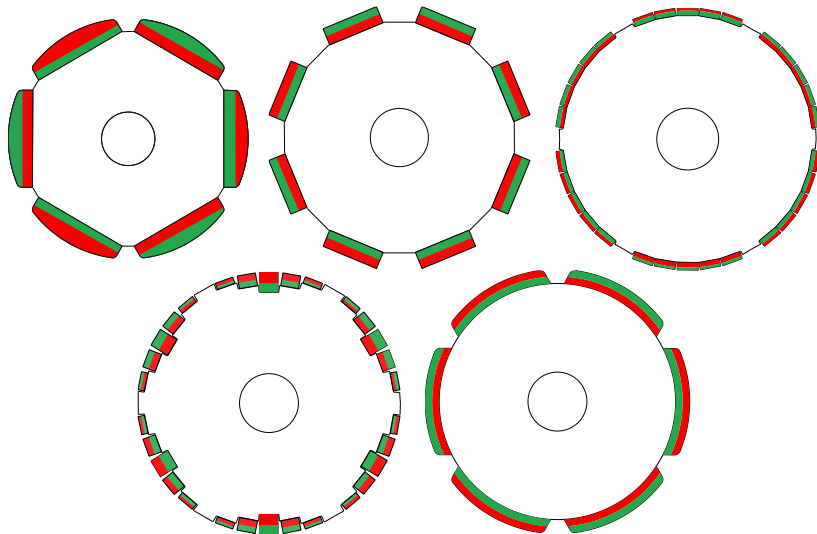


Graphic: Elektronik Praxis

The magnets of the permanent-magnet synchronous machine can be mounted on the rotor surface or inserted into internal cavities in the rotor.

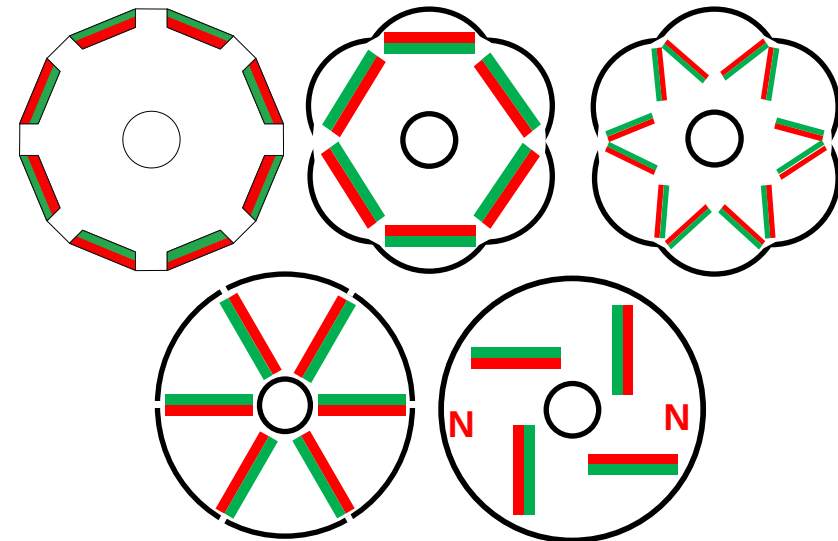
Surface Mounted Permanent Magnets - SPM

- Magnets are glued to the surface of the rotor sheet package
- To maintain rotor mechanical stability under high centripetal forces, a non-magnetic retaining sleeve is required for internal rotors
- SPMs can be dismantled and recycled more easily



Interior Permanent Magnet - IPM

- No gripper required for assembly, so it can be easily automated (e.g. by means of a joining module)
- Depending on the design, high assembly forces must be taken into account
- No retaining sleeve required
- → Air gap can be made smaller than SPM
- Dismantling of the magnets is more difficult



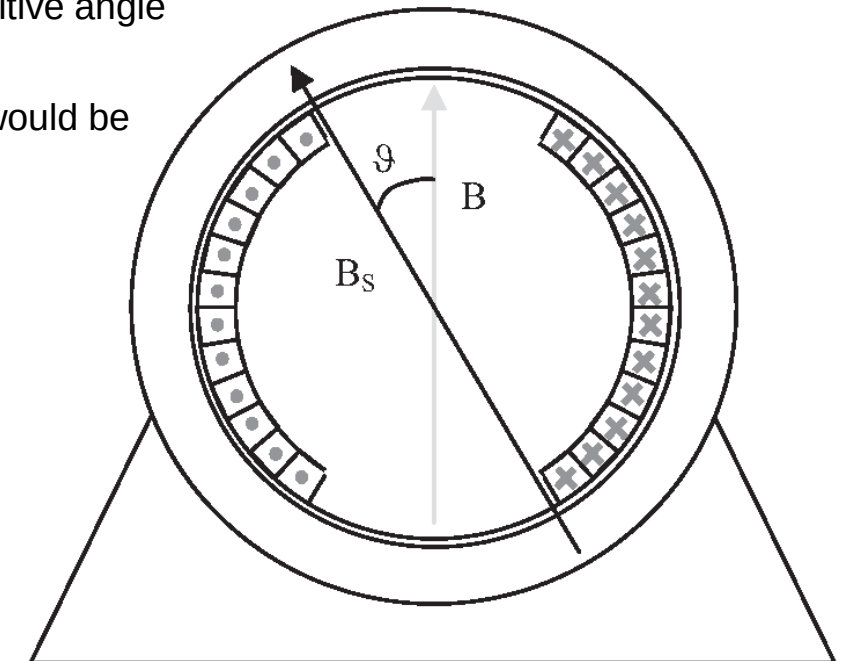
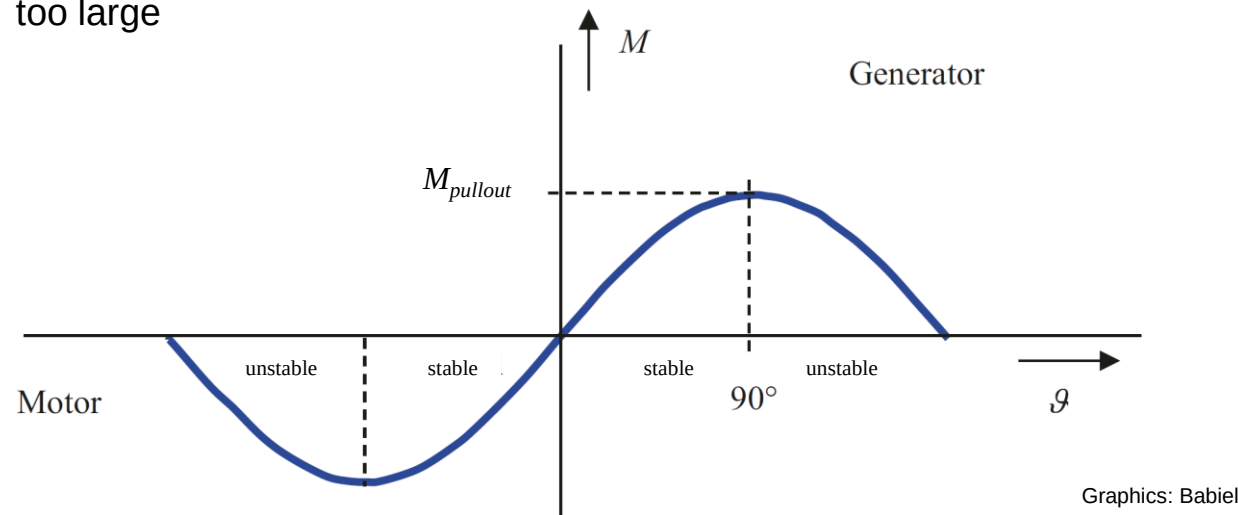
The speed response of the synchronous machine depends on the rotational frequency of the excitation field.

- The synchronous machine rotates at exactly the same frequency as the excitation field
- Under load, the torque angle ϑ changes (angular difference between rotor field B and stator field B_s)
- As ϑ increases, more torque is supplied by the rotor
- The synchronous machine can theoretically be operated in a range of $-90^\circ < \vartheta < 90^\circ$
- In practice, however, the range is limited to $-70^\circ < \vartheta < 70^\circ$
- A negative torque angle occurs when the machine is operating as a motor, a positive angle occurs when operating as a generator
- Starting under load is not possible at mains frequency, as the pole wheel angle would be too large

$$n_s = \frac{f}{p}$$

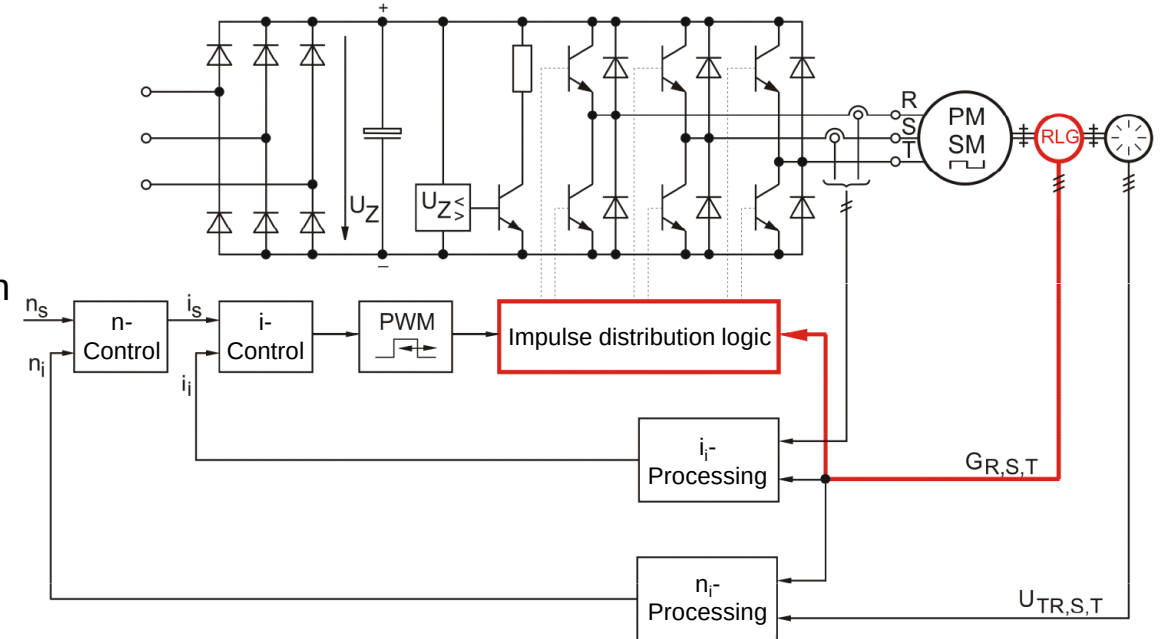
mit:

- n_s – Synchronous speed
- f – Frequency
- p – Number of pole pairs



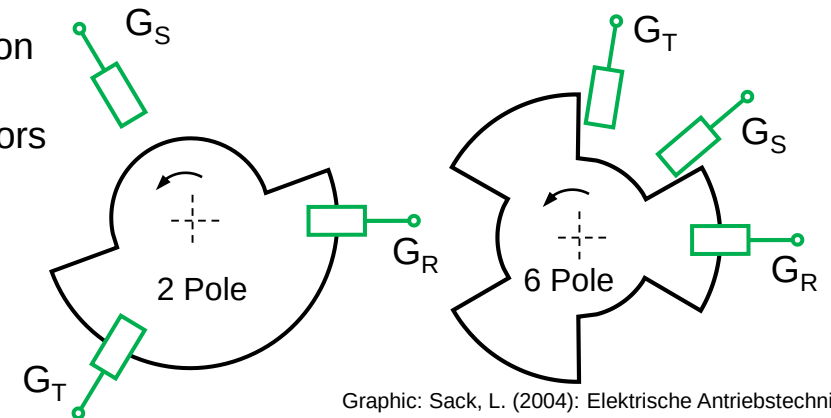
Block commutation is a relatively simple way of controlling synchronous machines.

- Rotor position is determined
- Rotational speed is calculated
- Transistor bridges energize the respective phase
- Current and voltage are regulated
- When operating on a three-phase network, bridge rectification of the supply voltage is carried out
- The pulsed DC voltage is stabilized and smoothed by the DC link capacitor
- The transistors are powered by DC voltage



Another option is room indicator modulation.

Simple rotor position determination with photoelectric sensors 2-pole (left) and 6-pole (right)



Graphic: Sack, L. (2004): Elektrische Antriebstechnik

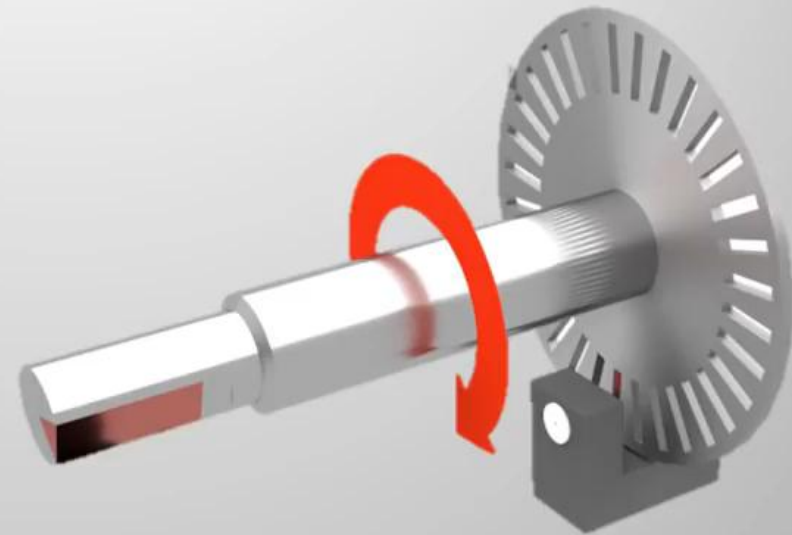
By counting pulses, incremental encoders can record the change in position of the electric drive and give this back to monitor precise control tasks.

Process

Incremental rotor encoder

Video:
Learnchannel

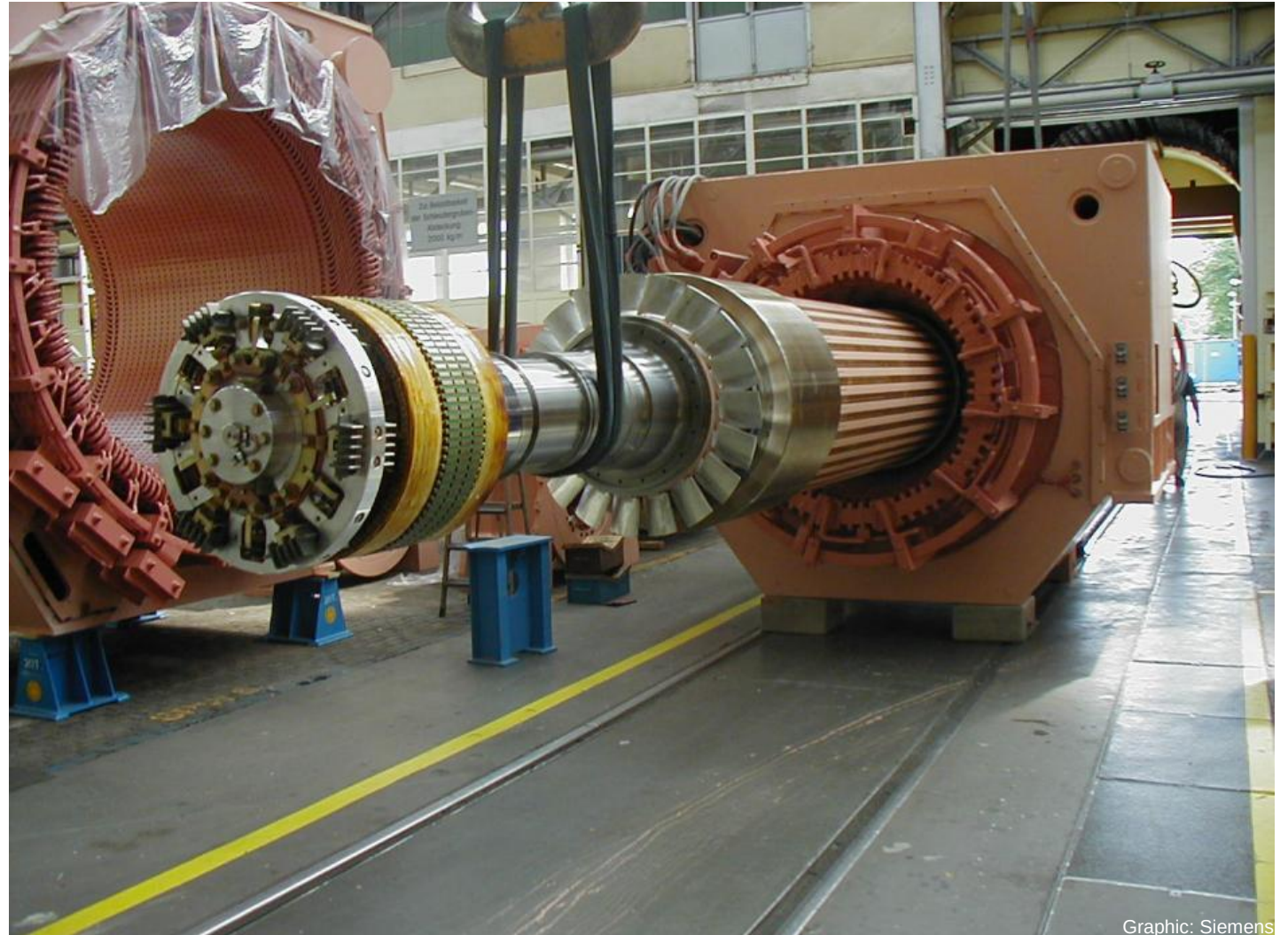
<https://www.youtube.com/watch?v=zzHcsJDV3>



YouTube Learnchannel-TV

In Hammerfest (Norway), 65 MW synchronous motors are used for gas liquefaction.

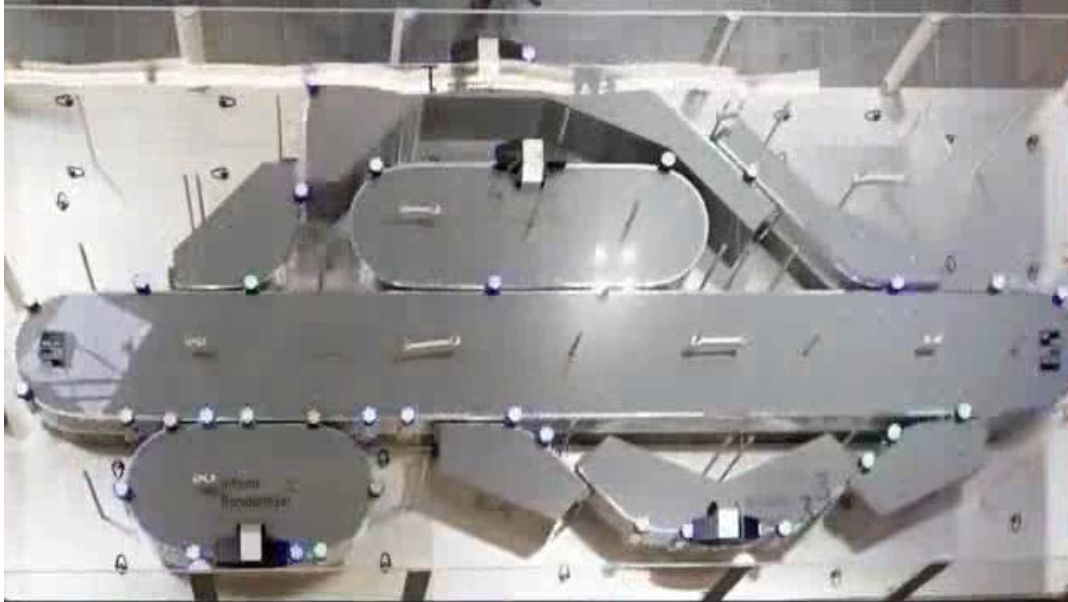
- Power: 65 MW
- RPM 3600 min⁻¹
- Inductively excited rotor
- Engines are among the most powerful in the world
- Generators in power plants output up to over 1000 MW



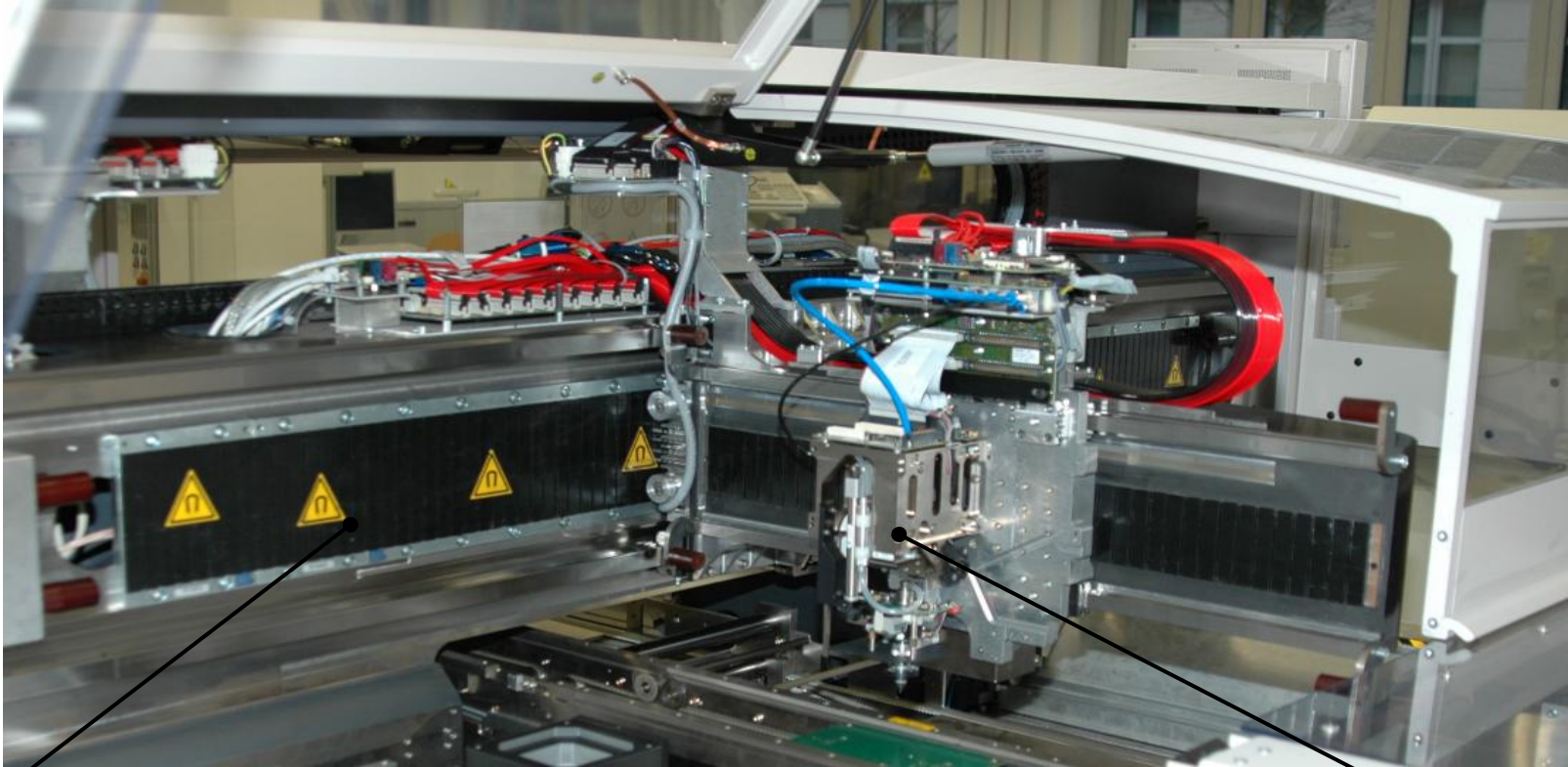
Permanent magnet linear motors have a significantly improved operating characteristics compared to conventional spindle drives.

Advantages of linear motors

- High acceleration
- Very good controllability
- Low accelerated masses
- High positioning accuracy due to short kinematic chain
- Simple mechanical design
- Drive components (“rotors”) can be exchanged during operation
- Able to integrate track switches



The use of linear motors in pick-and-place machines makes it possible to achieve high output with increased placement accuracy.



Y-Positioning

max. acceleration: 3 g
max. travel: 1500 mm
accuracy: 3 μm

X-Positioning

max. acceleration: 3.5 g
max. travel: 450 mm
accuracy: 3 μm

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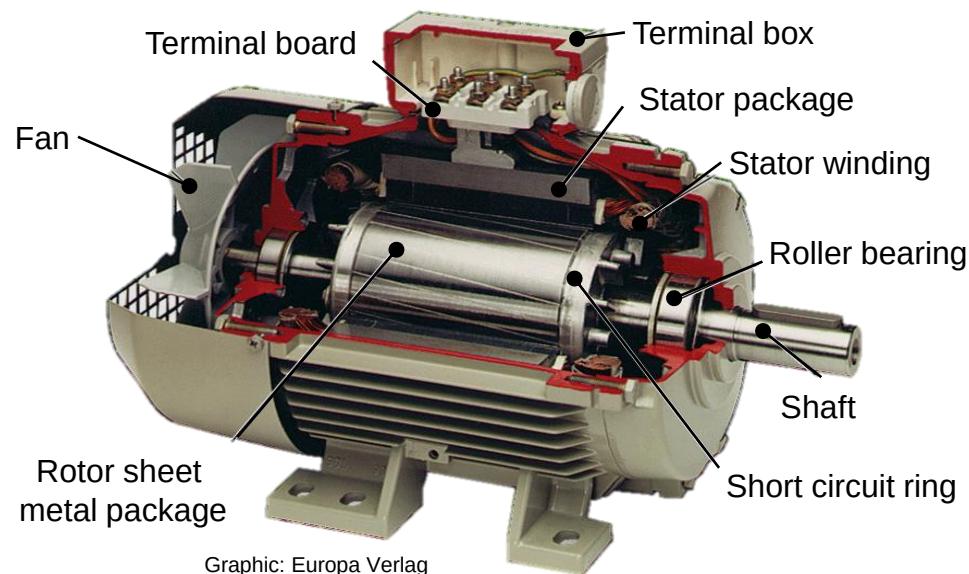
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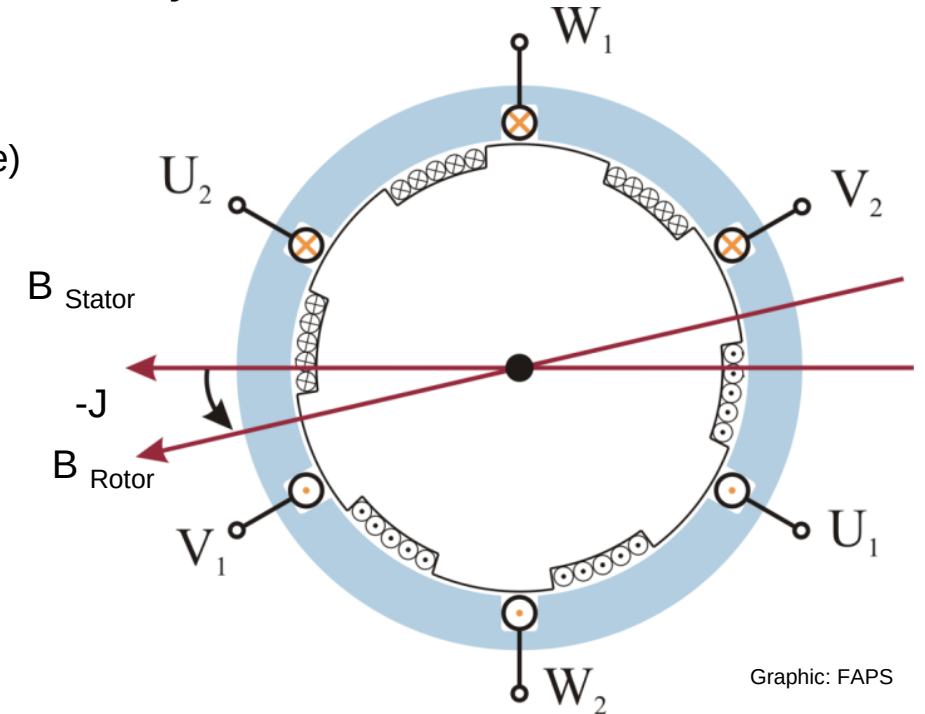
Graphic: BMW

Asynchronous machines use the same stator as synchronous machines.

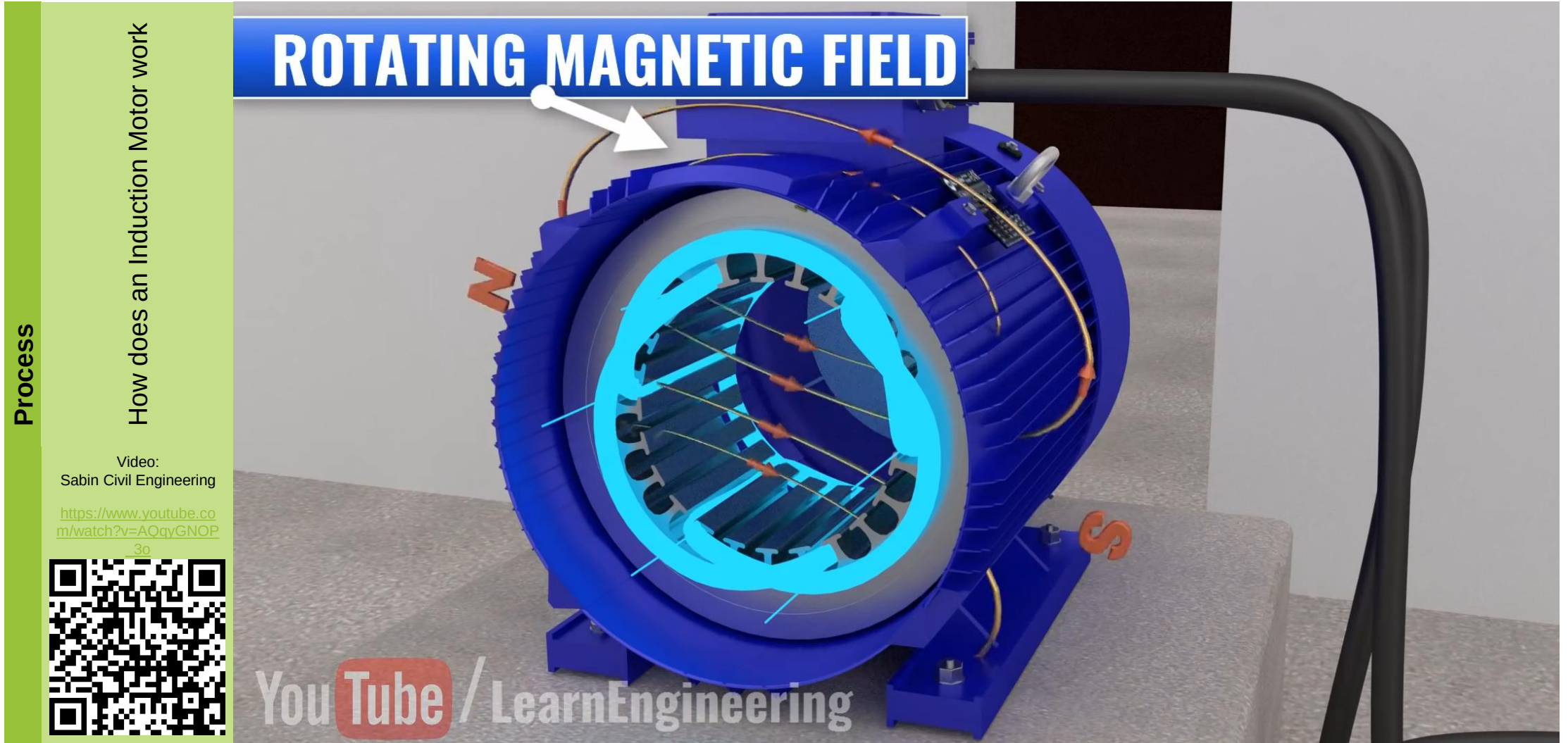
- Windings are housed in the stator
- The stator windings are each offset by 120°
- The rotor does not have permanent magnets, but short-circuited coils
- The rotor windings (inductors) are often constructed with shorting bars (cage)
- Stator and rotor are made of laminated sheet metal to reduce eddy currents
- The asynchronous machine can be operated in wye or delta mode



Asynchronous machine with wound rotor

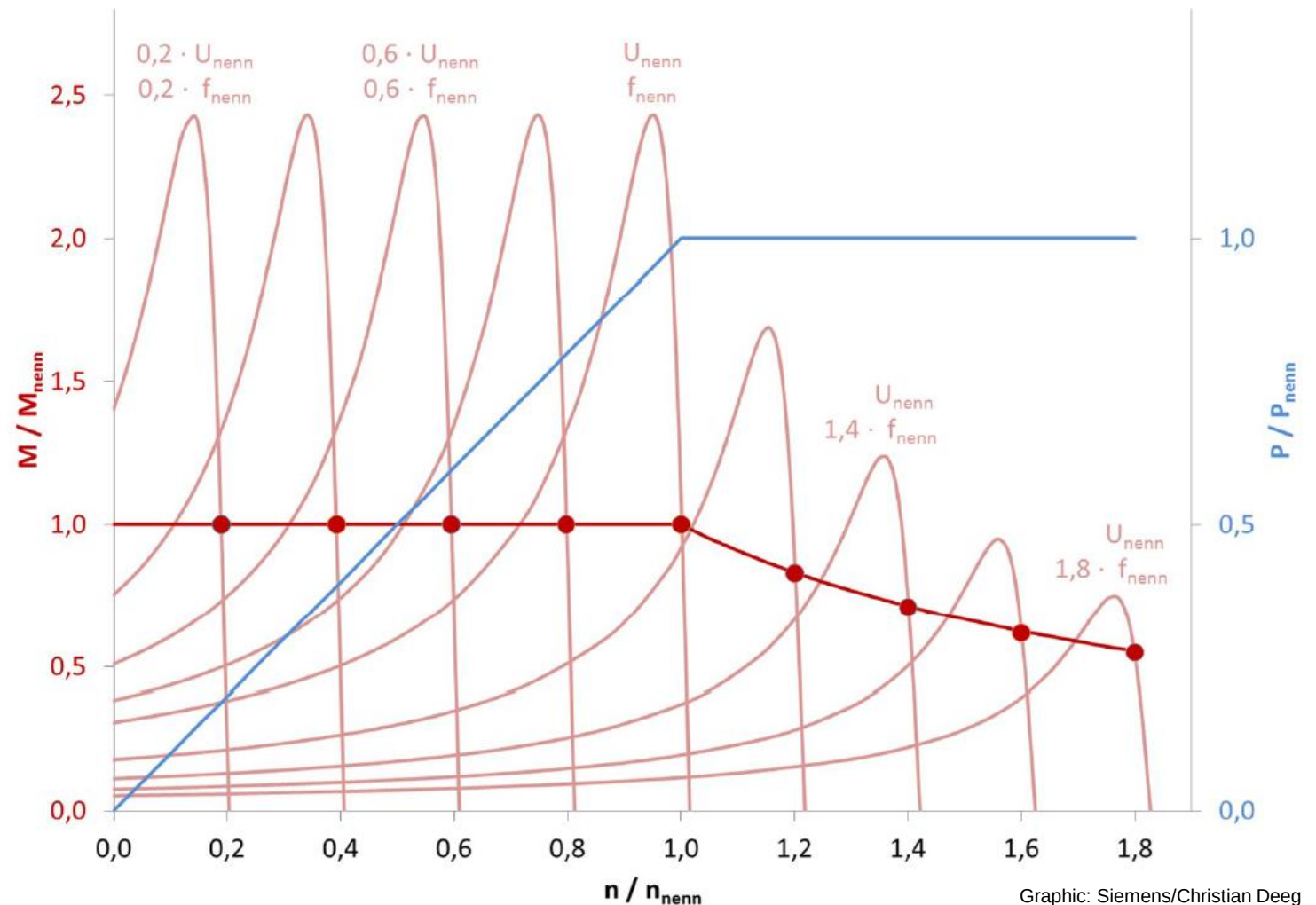


Asynchronous machines use the same stator as synchronous machines.



By operating the asynchronous machine with a converter, a different operating behavior is forced on the machine.

- The converter characteristic curve is made up of a group of characteristic curves.
- The characteristic curve can be designed as desired within the limits of the machine.
- In U/f mode, both frequency and voltage are varied.



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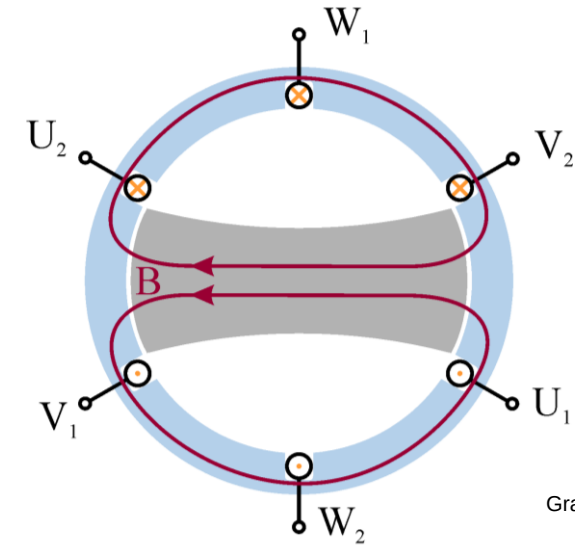
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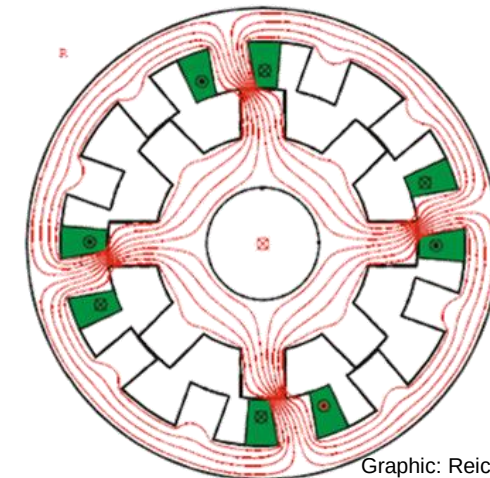
Graphic: BMW

Reluctance motors work according to the principle of minimal reluctance (magnet. resistance).

- Operating behavior similar to that of the synchronous machine
- Stator design comparable to synchronous motor
- Rotor is made of ferromagnetic material with high permeability (without permanent magnets)
- The rotor has distinct poles and rotates in the direction of minimum magnetic resistance for the stator magnetic field
- Design allows the realization of cost-effective synchronous reluctance machines
- Robust, cost-effective design (no magnets)
- Suitable for high speeds
- Less efficient than a synchronous machine, but still high
- Stator currents are higher compared to synchronous machines, as additional magnetization of the lamination stack is required
- The principle is often used for stepper motors, whereby the stator field is not continuously rotating, but switched step by step

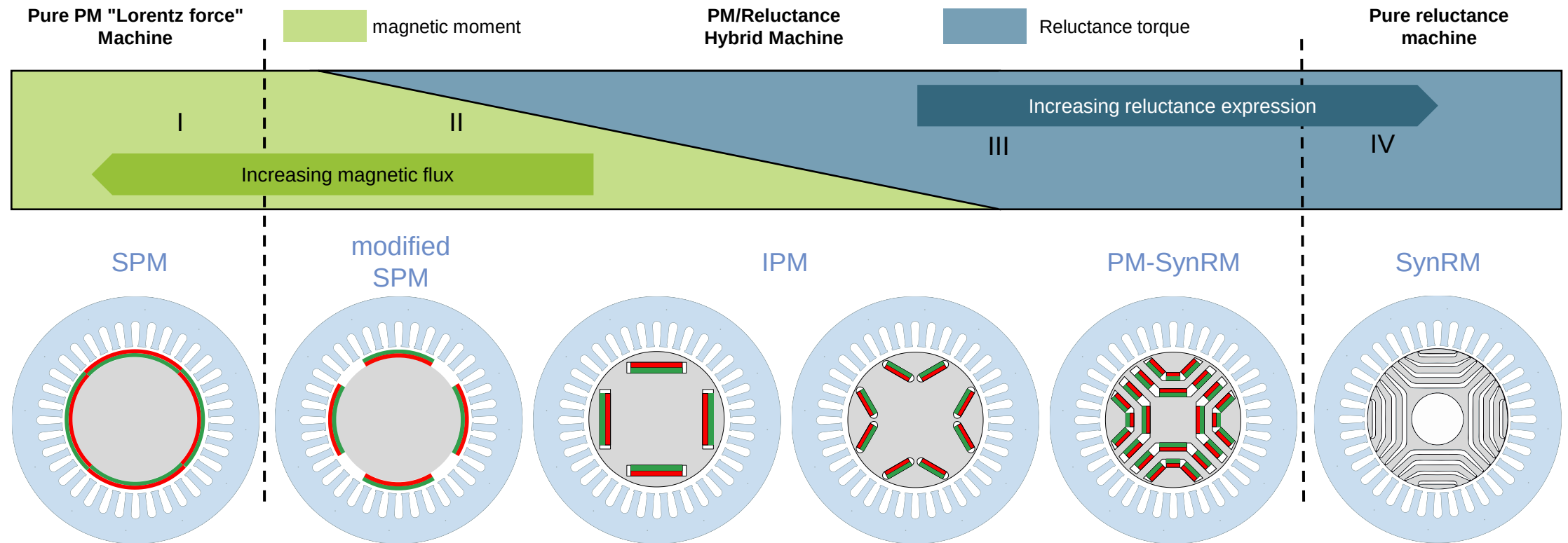


Graphic: FAPS



Graphic: Reichert, ETH Zürich

A hybrid of synchronous and reluctance machine can be constructed by selecting the appropriate magnet arrangement and sheet metal cutting geometry.



Based on
Morimoto (IEEE 2007)

The transverse flux machine has both windings and current flow parallel to the direction of rotor motion.

- Force is generated by generating minimal magnetic resistance (reluctance) (also PM supported)
- The linear actuator shown can also be designed as a rotary drive
- Actuators offset by several fractions of a pole pitch are always necessary to generate an arbitrary force vector (moment)

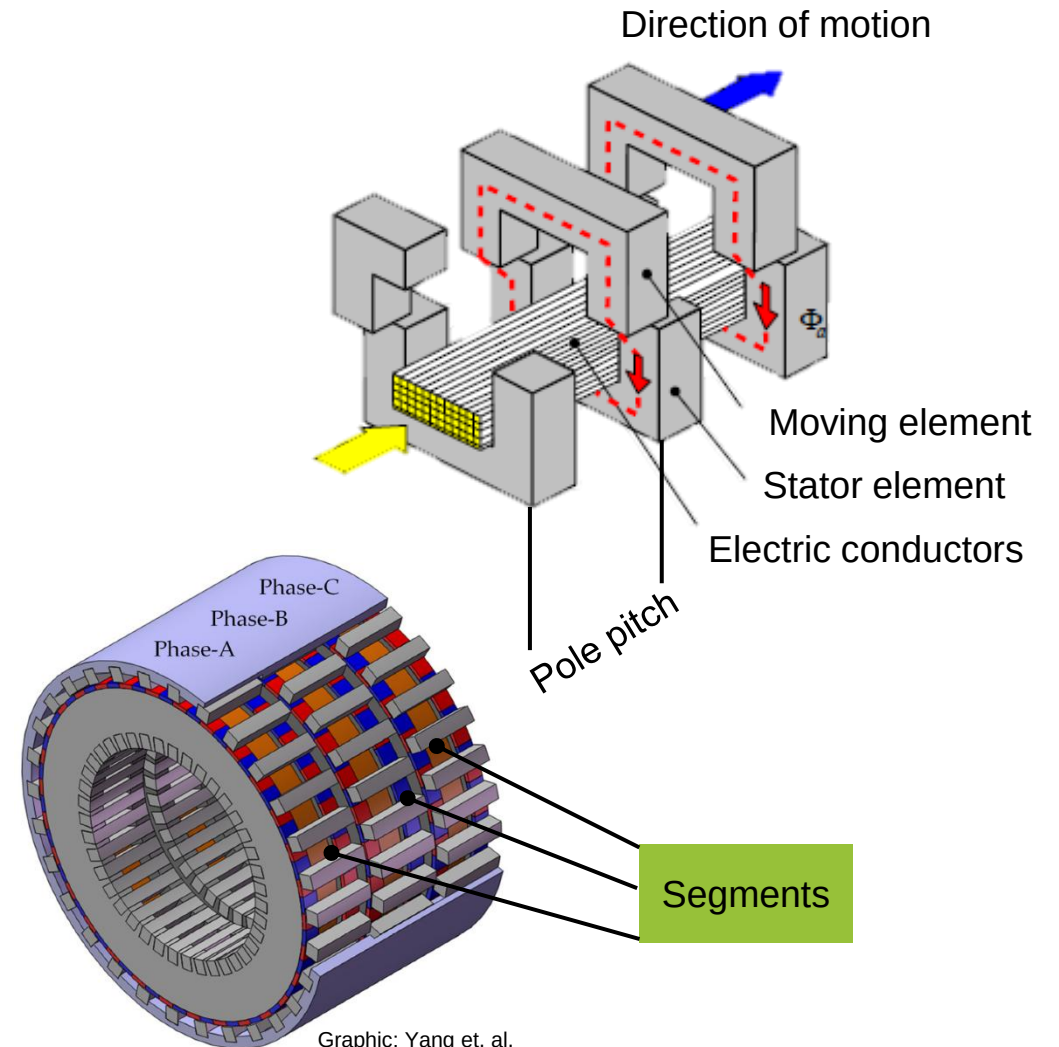
Comparison of transverse flux machine and (conventional) longitudinal flux machine:

Advantages

Ring-shaped winding →
No winding heads, lower losses, higher torque density

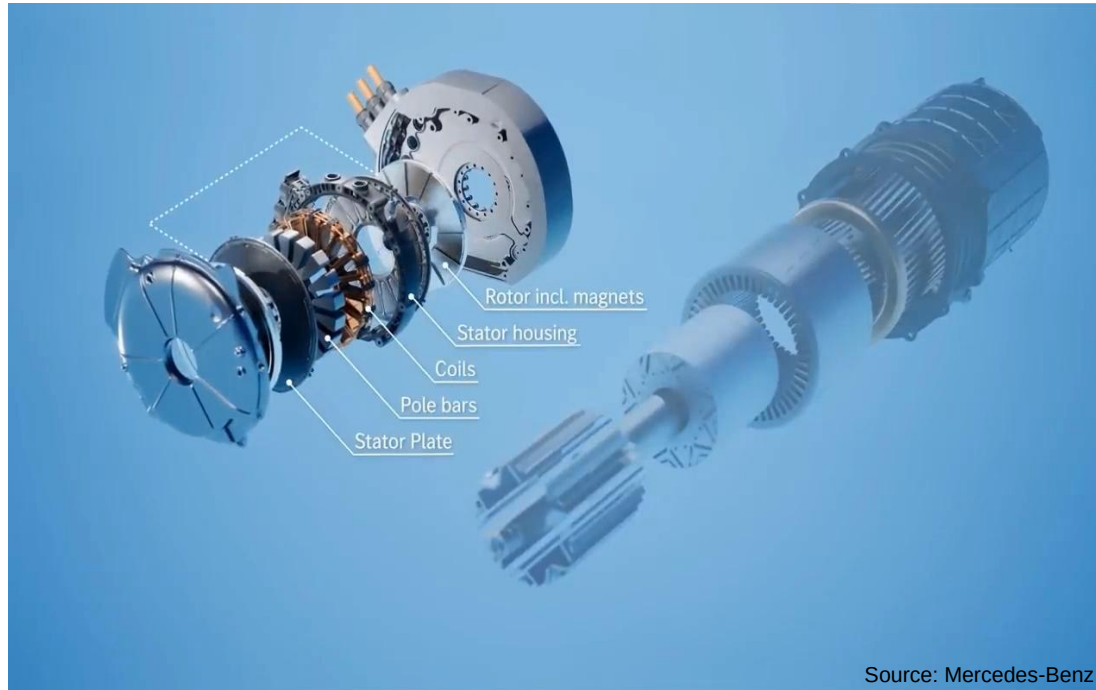
Disadvantages

- High torque and normal force fluctuations
- Complex control and regulation required
- High reactive current



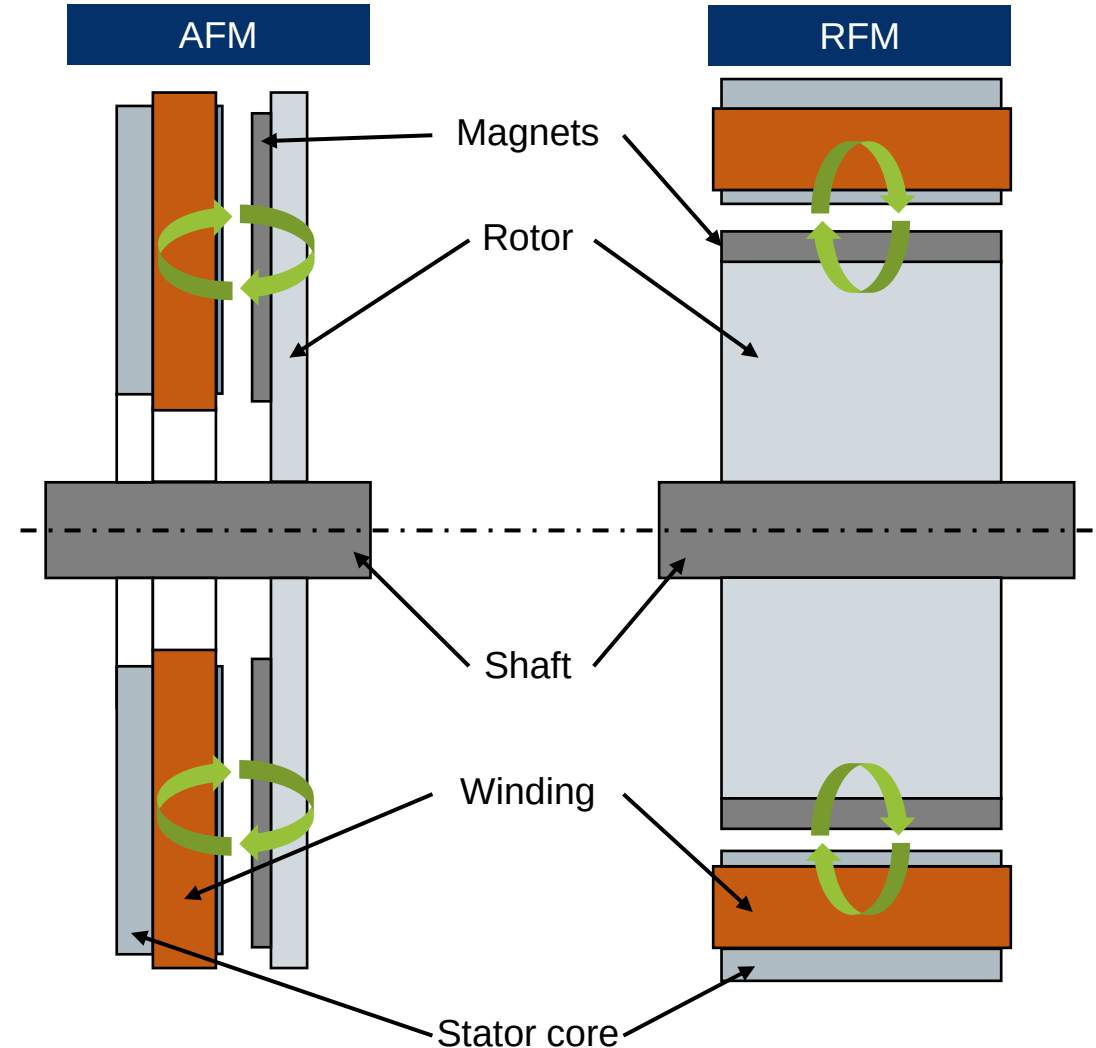
Graphic: Yang et. al.

Axial and radial flux motors differ in their basic geometric design due to the different propagation direction of the magnetic field.

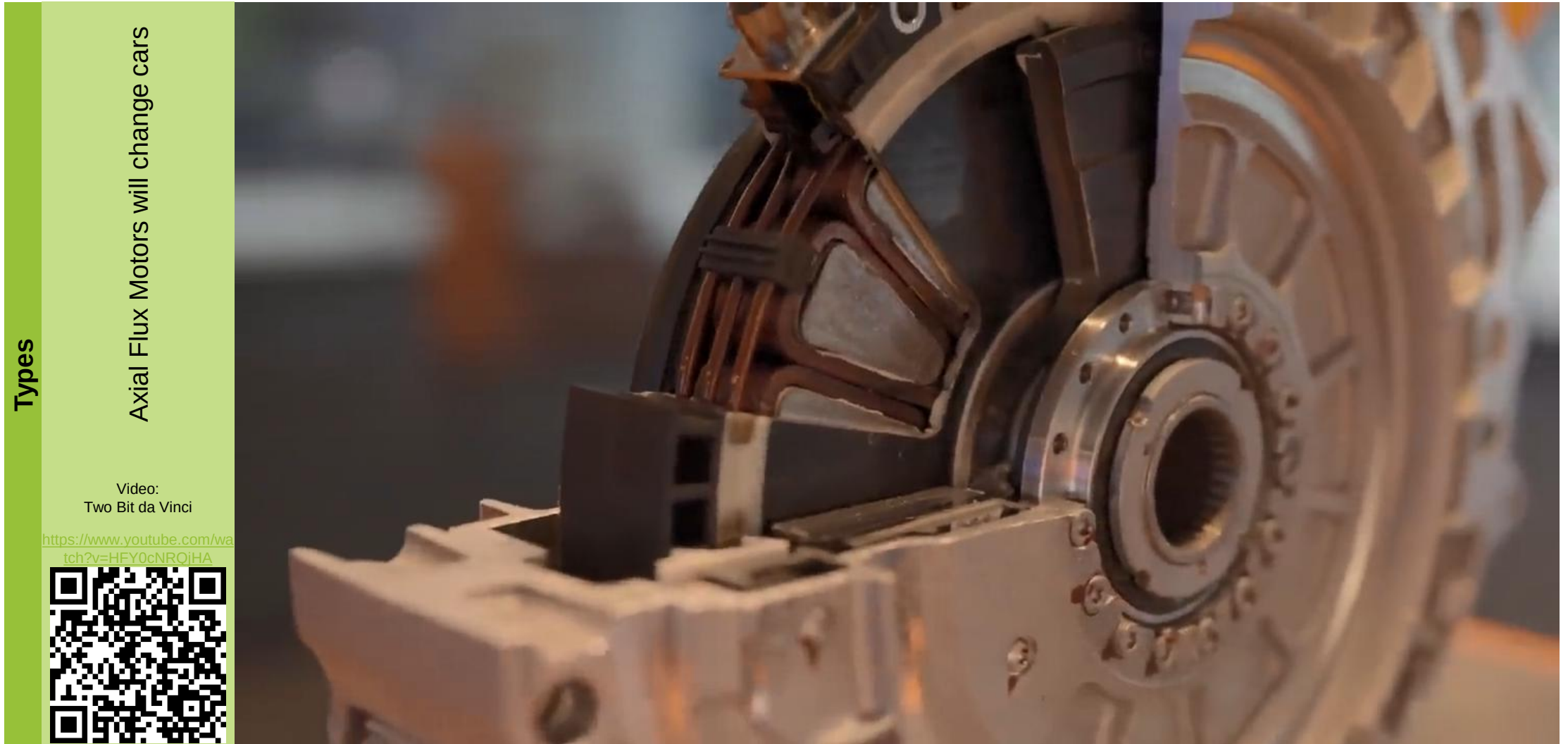


Reasons for the usage of AFM

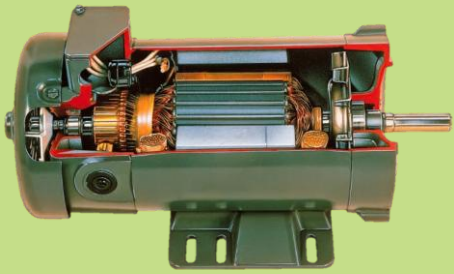
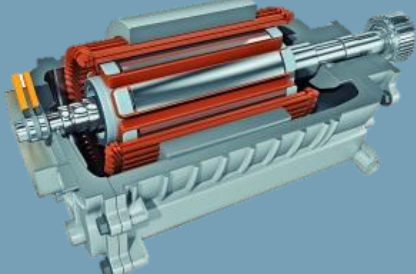
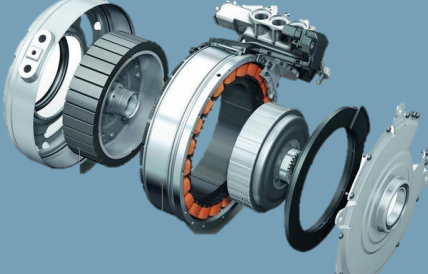
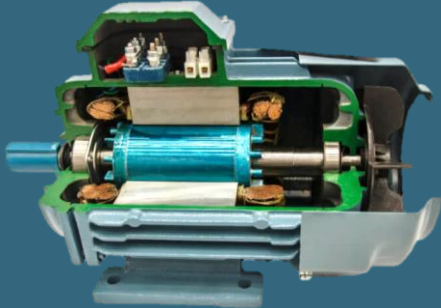
- Higher power and torque densities
- Reduced losses
- Compact design
- Reduced weight



A disc motor or axial flux motor is an electric motor whose rotor has the shape of a disc.



Comparison between DC, synchronous, and asynchronous machines shows that each is suitable for different applications.

	mechanically commutated DC machine	Synchronous machine		Asynchronous machine
		externally excited	permanently excited	
Abbreviation (DE)	DCM (DE: GSM)	EE SM (DE: FE SM)	PE SM	ASM
Production costs	low	medium	high	medium
Maintenance required	high	high	low	low
Power-to-weight ratio	low	high	very high	medium
Efficiency	medium	good	very good	medium
Example				
	Graphic: Courtesy Reliance Electric Co.	Graphic: Continental	Graphic: Audi	Graphic: Günter Pauli GmbH

Complete overview of different electric motor types.

Types

Electric Motor Types and Complete Overview

Video:
OSWOS

<https://www.youtube.com/watch?v=0YtiH0gPPcY>



ELECTRIC MOTORS

COMPLETE OVERVIEW OF DIFFERENT TYPES

The first lecture unit includes a general overview of electromechanical engineering and an introduction to the associated research area.

- Organizational matters
- Motivation and definition of terms
- Structure of an electrical machine
- Fundamentals of electromechanical energy conversion
- Structure of the fields of application
- Types, operation, and control systems of various electrical machines
 - Transformers
 - Commutator machines
 - Synchronous machines
 - Asynchronous machines
 - Other designs
- **Current research projects**



Graphic: BMW

The E|Form project aims to highly automate the production of electric motors with formed coil technology in rail-based heavy-duty transport.



E|Form

Three-stage, automated end forming of flat conductor coils in a laminated core

Project budget:

Total: 1,42 Mio. €

FAPS: 720.000 €



DLR Projektträger

A key approach in the E|Form project is to automate the production of electric motors with preformed coil technology at Siemens Mobility. A total three-stage assembly environment is to be created through the use of sensors and actuators as well as devices specially designed for the assembly process. With its help, a significant reduction in assembly time is possible, since the main assembly of the pre-wound coils is not sequential and manual, but comprehensively automated.

FAPS

Automation

Development of the coil geometry

- Conductor dimensions/material
- Primary insulation
- Forming process

Development of automation concepts

- Digital process monitoring
- Coil feeding
- Coil detection



SMO

Assembly

Product and process engineering

- Specifications
- Industrialization

Prototype engine assembly

- Testing
- Electrical
- Mechanical

SPS



WEBO

Engineering

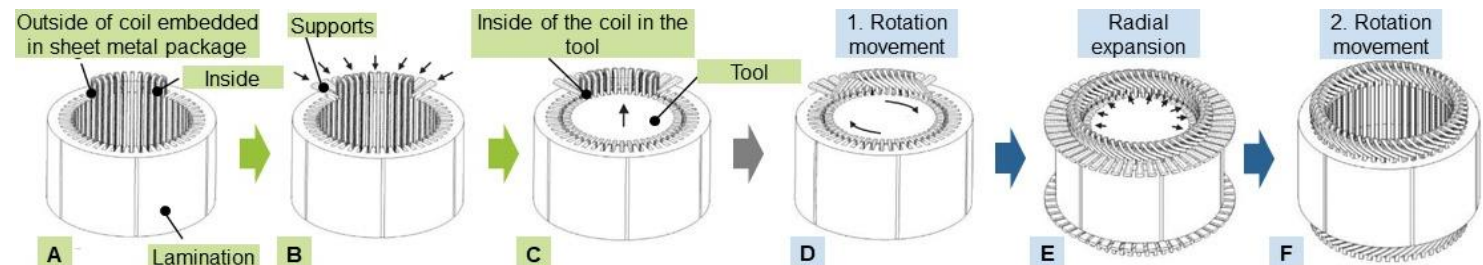
Process engineering

Construction of prototypical sub-processes

- Rotational forming
- Expansion forming
- Consolidation of process steps into overall structure



Three-stage, automated end forming of flat conductor coils in a laminated core



FAPS

SIEMENS

WEBO

Project duration:

01.09.2023 to 31.09.2026

The FAR project aims to develop highly automated manufacturing processes for the production of wheel hub motors for electric vehicles and commercial vehicles.



FAR – Development of novel manufacturing processes for a highly efficient drive in the wheel hub

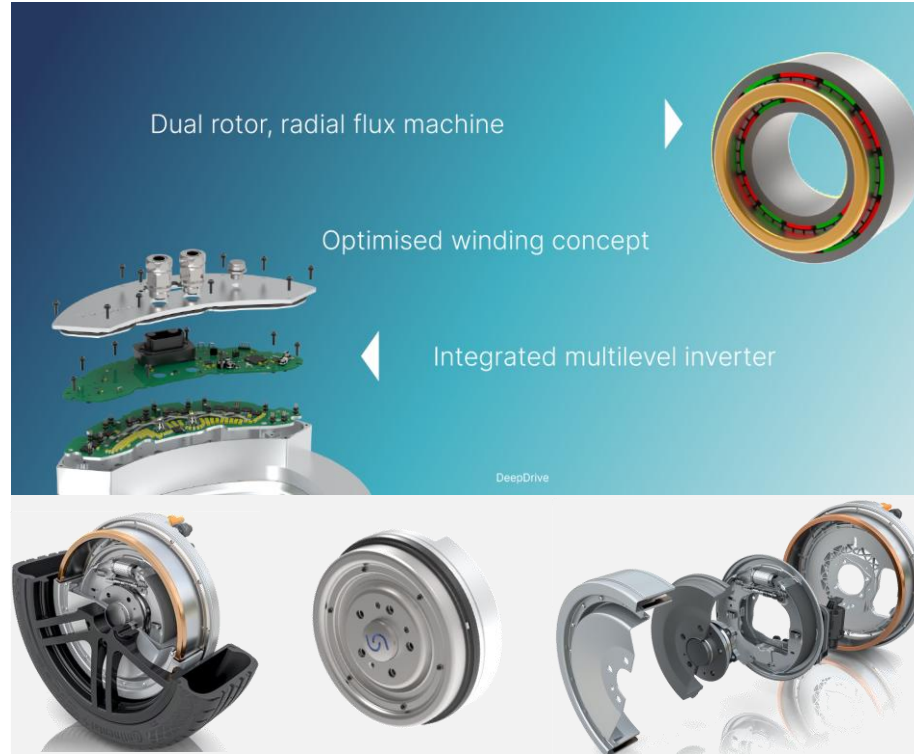
bayern  innovativ

Bayerisches Staatsministerium für
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The research project addresses the development of automated manufacturing processes for the production of highly efficient wheel hub motors for electric vehicles and commercial vehicles with regard to large-scale production.

 **DeepDrive**

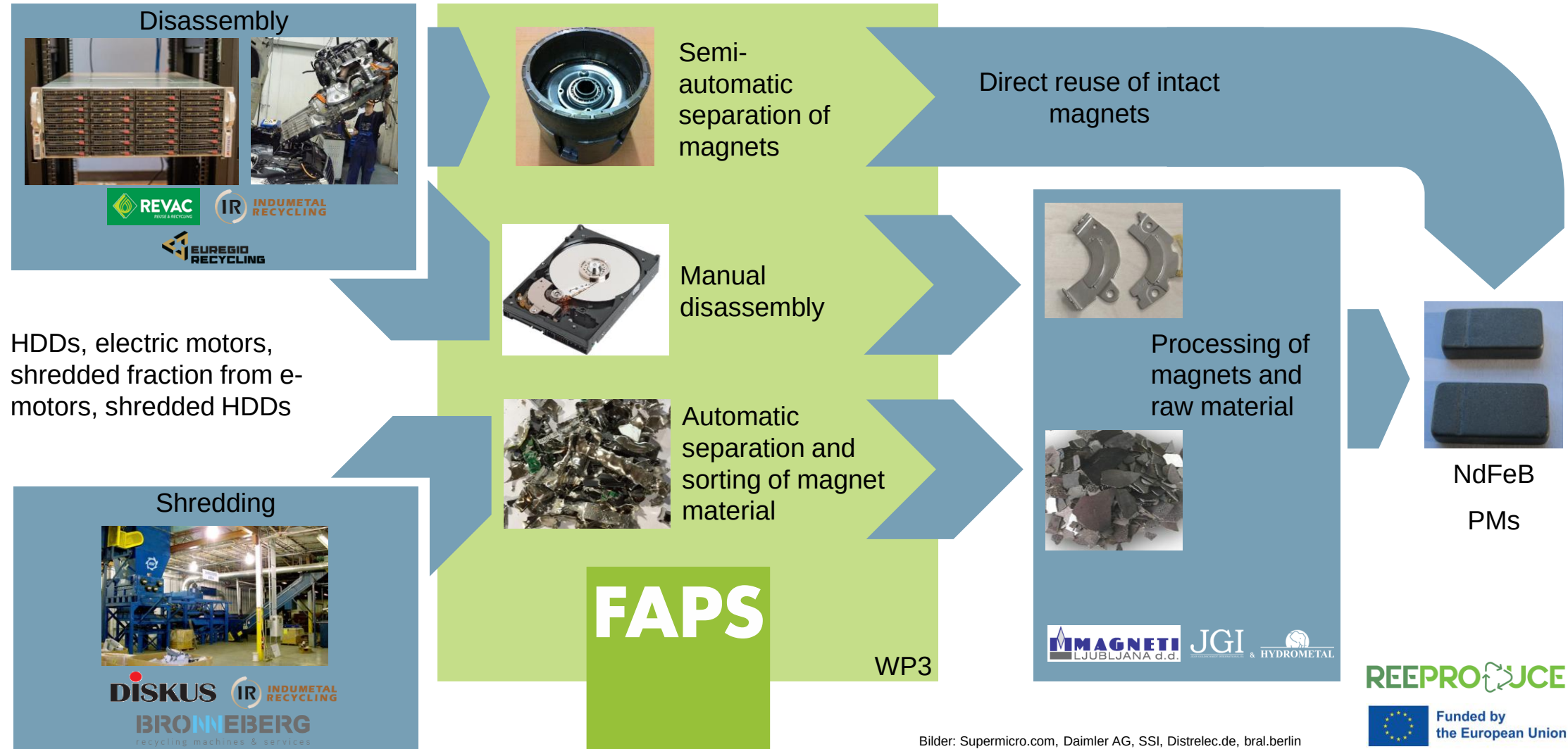
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Key areas of work for FAPS:

- Process development for the production of the winding bars according to the defined requirements
- Transfer of the developed stator winding to the joining process
- Process development of an economical and automatable joining process for the production of the stator winding and the interconnection module
- Verification and validation of the developed winding by means of quality testing

The goal of the research project REEPRODUCE is the economic separation and reuse of rare earth element containing permanent magnet material from WEEE.



Bilder: Supermicro.com, Daimler AG, SSI, Distrelec.de, bral.berlin

REEPRODUCE

Funded by
the European Union

The aim of the project E|KI-Opt is to exploit the potentials of artificial intelligence to optimize the production of brushless DC motors.



Explainable artificial intelligence for the holistic optimization of processes in electric motor production

The aim of the project E|KI-Opt is to exploit the potentials of artificial intelligence (AI) for process and production optimization in electric motor manufacturing. In addition to considering the entire process chain, essential sub-process steps, such as winding, contacting and pressing, are to be optimized using AI. A special focus is put on the explainability of the developed AI models.

Partners:

- Bühler Motor GmbH
- DATA AHEAD AG
- Lingl Schweißtechnik GmbH
- Institute FAPS of the Friedrich-Alexander-University Erlangen-Nuremberg (FAU)

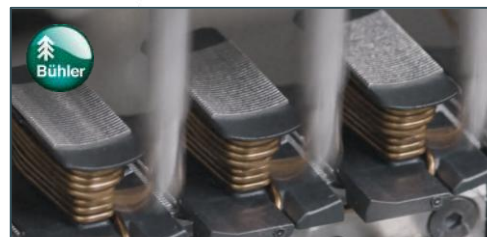
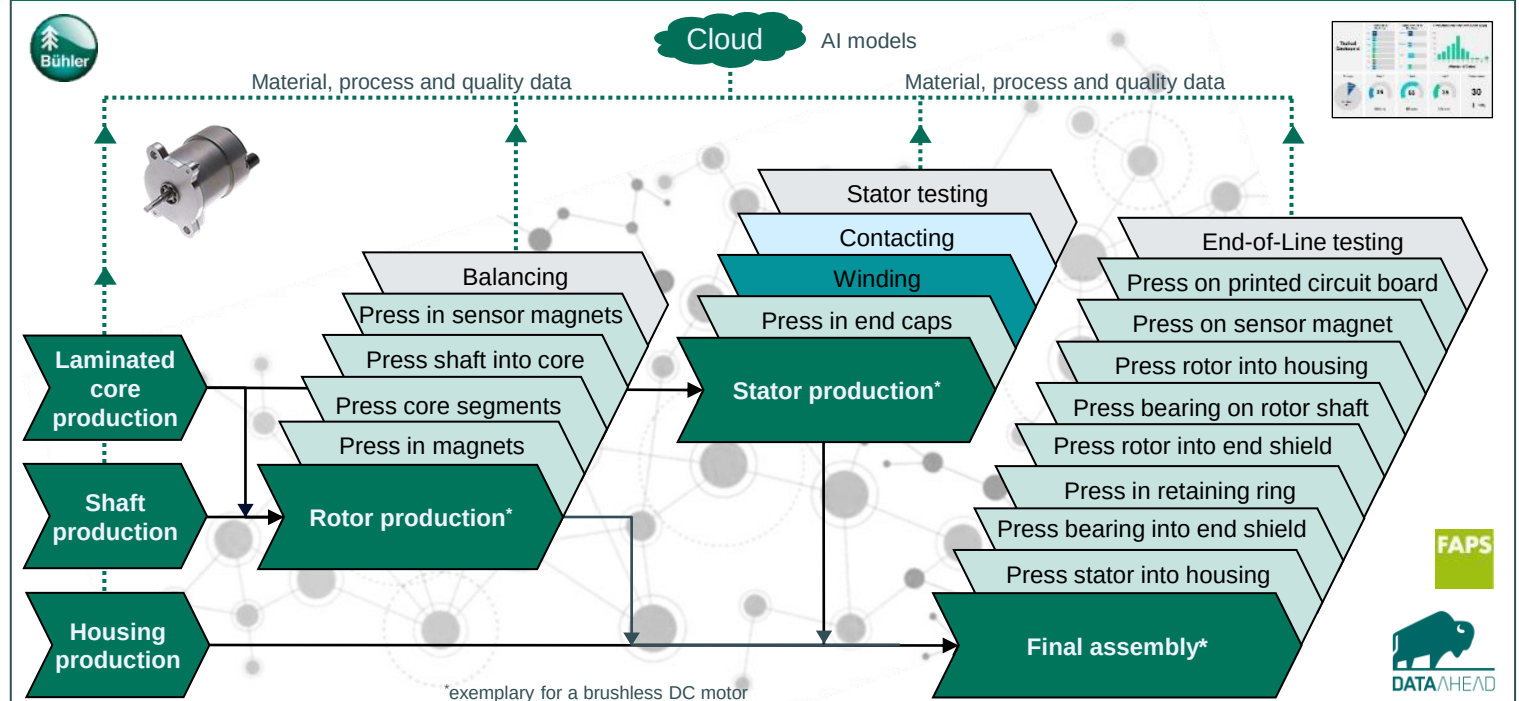
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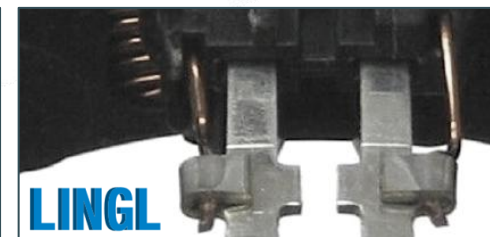
Bayerisches Staatsministerium für
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ZD.B ZENTRUM
DIGITALISIERUNG
BAYERN

Use Case 1: AI-based detection of cross-process correlations in electric motor production



Use Case 2: AI-based optimization of winding processes



Use Case 3: AI-based optimization of contacting processes



Use Case 4: AI-based optimization of press-on and press-in processes

The aim of the EIMPOWER research project is to develop a fully industrialized and digitized process chain for the construction and operation of inductive ERS.



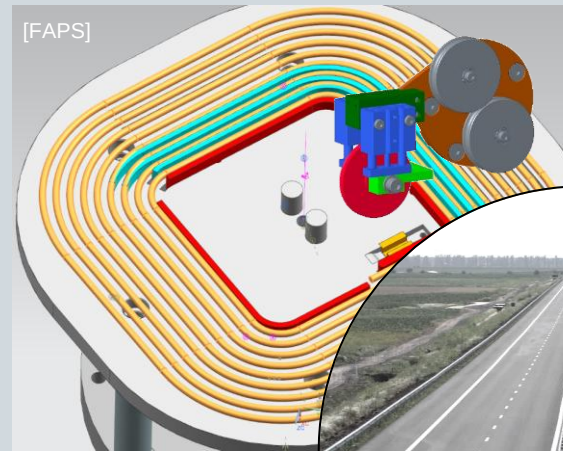
Automated manufacturing processes for Electric Road Systems for the electrification of heavy-duty traffic

Project sum:
ca. 8,00 Mio. €

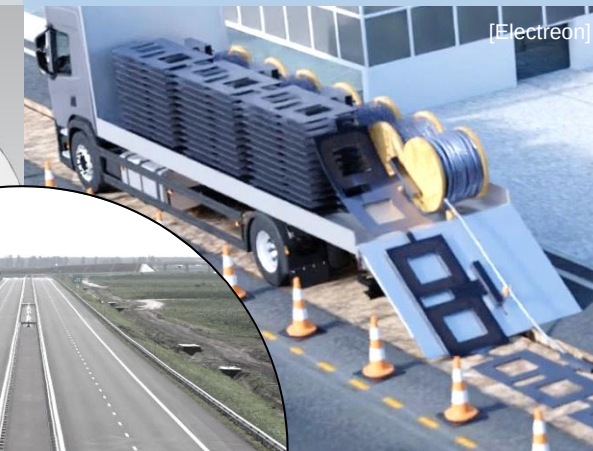
The aim is to open up the entire value chain for the construction of an electrified road (ERS) on the process side and to record all relevant processes from the production of the coils to the operation of the road across interfaces by means of innovative measurement methods and digital approaches.

2022-2025

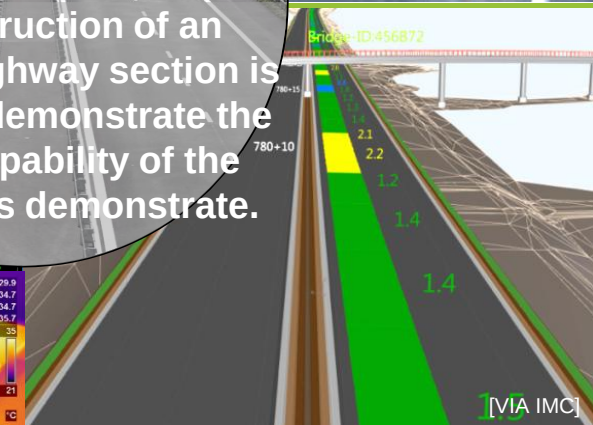
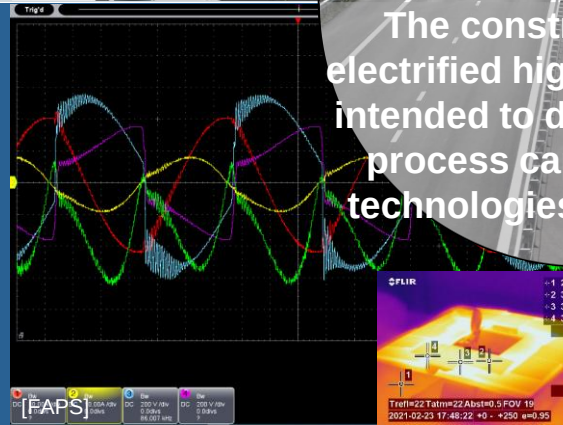
Processes for manufacturing coil systems



Processes for building inductive ERS



The construction of an electrified highway section is intended to demonstrate the process capability of the technologies demonstrate.



ERS quality monitoring

Digitization of the process chain

A large number of student projects are advertised in the chair's research area of electrical engineering, which transfer what has been learned into practice in the long term.

StudOn

https://www.studon.fau.de/studon/goto.php?target=cat_164867



faps.de

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Elektromaschinenbau / EDrive-Center

Das EDrive-Center (Bayerisches Technologiezentrum für elektrische Antriebstechnik) bietet vielfältige und praxisnahe Themen speziell für elektrische Antriebe und kontaktlose Leuchtsysteme, u.a. in den Bereichen Fertigungs- und Prüfprozesse, Versuchsanleitung- und -durchführung, robotergestützte Montageprozesse, energie- und ressourceneffiziente Produktion sowie Künstliche Intelligenz und Maschinelles Lernen.

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E/Drive-Center
Bayerisches Technologiezentrum für elektrische Antriebstechnik

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Dateien

- HIWI_Unterstützung der Forschungstätigkeiten im Bereich des hybrid-elektrischen Fliegenreibring
Andreas Reidel, Themengruppe Elektrischer Antrieb
pdf 134,1 KB 09. Jul 2021
- BA_PA_MA_Unterstützung der Forschungstätigkeiten im Bereich des hybrid-elektrischen Fliegenreibring
Andreas Reidel, Themengruppe Elektrischer Antrieb
pdf 51,0 KB 09. Jul 2021
- BA_PA_MA_Konstruktion und Aufbau einer Prüfvorrichtung für die Bestimmung von weichen magnetischen Kenngrößen in Traktionsantrieben
Andreas Reidel, Themengruppe Elektrischer Antrieb
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FAPS Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik Prof. Dr.-Ing. Jörg Franke

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- Elektromaschinenbau**
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Konzeption und Implementierung der Simulationskomponente eines Demonstrators zur Planung von Produktionssystemen mittels des Digitalen Zwillings

Ausgangslage: Der Einsatz der Simulation ist in der Planung von Produktionssystemen weit verbreitet. Eine frühe Planungsaufgabe stellt die Auslegung des

The contents of the lecture unit can be further deepened with the following specialized literature:

Textbooks

- Bălă, C., A. Fetița und V. Lefter, Hg. *Handbuch der Wickeltechnik elektrischer Maschinen. Theorie, Konstruktion und Technologie von Wicklungen rotierender elektrischer Maschinen.* 2., stark bearb. Aufl. Berlin: Technik, 1976
- Fischer, R. *Elektrische Maschinen.* 17., aktualisierte Auflage. München: Carl Hanser Verlag, 2017. Hanser-Fachbuch. ISBN 978-3-446-43767-8
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- SACK, L. Elektrische Antriebstechnik. In: H.-J. Gevatter und U. Grünhaupt, Hg. *Mess- und Automatisierungstechnik in der Produktionstechnik.* 2., vollst. überarb. Aufl. Berlin: Springer, 2004. ISBN 3540212078
- SCHRÖDER, D. *Elektrische Antriebe – Grundlagen.* Berlin, Heidelberg: Springer Berlin Heidelberg, 2017. ISBN 9783662554470
- BÖHMER, E., D. EHRHARDT und W. OBERSCHELP. Transformatoren und Übertrager. In: E. Böhmer, D. Ehrhardt und W. Oberschelp, Hg. *Elemente der angewandten Elektronik. Kompendium für Ausbildung und Beruf : mit umfangreichem Bauteilekatalog.* 17., korrigierte und verbesserte Auflage. Wiesbaden: Springer Vieweg, 2018. ISBN 9783834814968
- PLASSMANN, W. und D. SCHULZ. Energietechnik. In: W. Pläßmann und D. Schulz, Hg. *Handbuch Elektrotechnik.* 7., neu bearbeitete Auflage. Wiesbaden: Springer Vieweg, 2016. ISBN 9783834810212



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Textbooks

- BABIEL, G. Analoge Elektrische Maschinen. In: G. Babel, Hg. *Elektrische Antriebe in der Fahrzeugtechnik. Lehr- und Arbeitsbuch*. 4., verbesserte und erweiterte Auflage. Wiesbaden: Springer Vieweg, 2020. ISBN 9783658221751
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- BINDER, A. Elektrische Maschinen und Antriebe. Grundlagen, Betriebsverhalten. Berlin: Springer, 2012. VDI-Buch. ISBN 9783540718499
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- FISCHER, R. Elektrische Maschinen. 17., aktualisierte Auflage. München: Hanser, 2017. Hanser-Fachbuch. ISBN 9783446452954

FAU Lectures

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